

American

Conference

Modelica & fmi

Atlanta USA 2026

October 12-14



Georgia Institute
of Technology



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CONFERENCE CO-CHAIR

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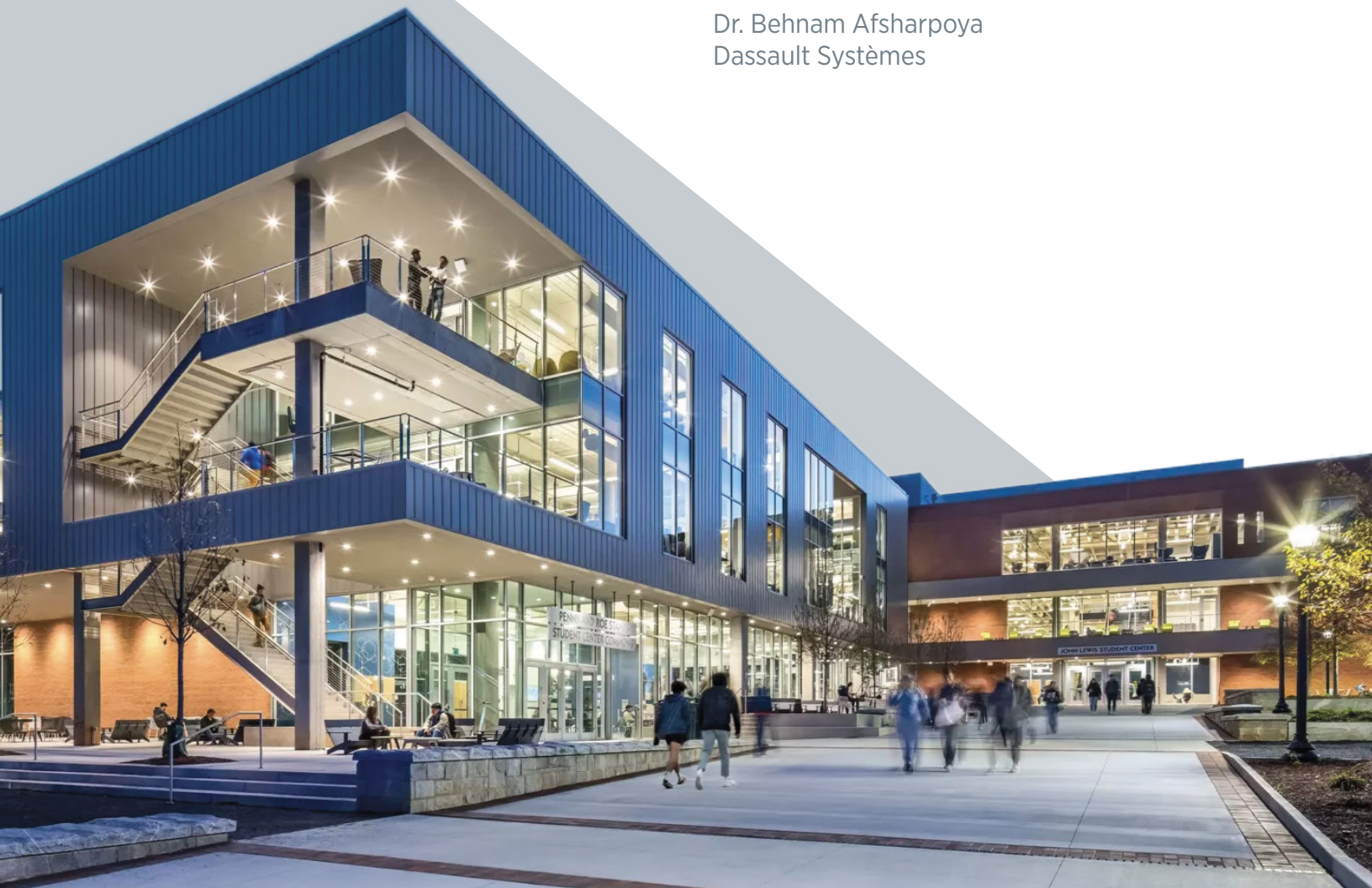
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WELCOME



Once again, we have been hard at work putting together another American Modelica Conference. This year's event will be taking place in Atlanta on October 12th-14th.

I spent my teenage years living in Texas and, like Georgia, we have some special pronouns which seem very appropriate here. Of course we have "you" like the rest of the country (second person singular). But in the south we have a different pronoun for second person plural (not unlike French or Spanish) and that pronoun is "y'all". So I might say that we, the organizing committee, would like to extend a warm welcome to y'all. But in the south we have another grammatical trick up our sleeves. You see, we have a special form of the second person plural pronoun indicating maximal inclusiveness and hospitality. And for that reason, I want to welcome all y'all to Atlanta for AMC 2026.

Obviously, the hot topic in 2026 is AI and agentic tooling. So you can expect to hear plenty about that. But one consequence of the explosion in AI capabilities is that models are increasingly untethered to specific languages and representations. For that reason and in the spirit of inclusiveness and southern hospitality, we decided this year to expand the call for papers to allow for a special track for Modelica and FMI adjacent technologies because the borders are rapidly blurring. We are eager to hear what all y'all think of this more inclusive format.

On a personal note, I'm excited to be having this year's event in Georgia and we are trying to introduce as many southern embellishments to the event as we can (for example, the conference dinner) and I hope you will enjoy the ambience of Atlanta while you are there.

See y'all soon.

Mike Tiller
Conference Co-Chair



The conference will take place at the Aerospace Systems Design Laboratory (ASDL) at Georgia Tech on Tuesday and Wednesday, October 13 and 14. Directions are available at https://www.asdl.gatech.edu/contact/visit_us/ or via other internet mapping services.

CONFERENCE ADDRESS:

Aerospace Systems Design Laboratory, 275 Ferst Drive NW, Atlanta, GA 30332.

Directions are available at https://www.asdl.gatech.edu/contact/visit_us/ or via other internet mapping services.

CONFERENCE PRESENTATIONS:

The presentations on Tuesday and Wednesday, October 13 and 14, will all take place on the second floor of the ASDL building.

CONFERENCE WEBSITE:

<https://modelica.org/events/american2026/>

All updated conference information can be found at the website, including conference papers and abstracts.

WORKSHOP ADDRESS:

The workshops will take place at the ASDL and the Student Success Center (SSC), which is located a short walk from the ASDL. The address of the SSC is 225 North Ave NW, Atlanta, GA 30332.

EMERGENCY INFORMATION:

To report an emergency, call 9-1-1, or call the Georgia Police Department at 404-894-2500 or from a campus landline at 4-2500.

INTERNET/WIFI:

The SSID is GTVisitor: if you are using a laptop, you may need to open a browser page first and attempt to navigate to a new website. Mobile devices should automatically bring up the login page or show a message stating you need to login. Once you open this page, please login via the free access capability.

REGISTRATION DESK:

The registration desk is located in the atrium on the second floor of the ASDL from Tuesday, October 13 at 7:30 am through the duration of the conference.

TOURIST INFORMATION:

Georgia Tech offers visitors a distinctive blend of historic landmarks, innovative research, arts, and green spaces in the heart of Midtown Atlanta. Conference attendees can explore the campus on a self-guided walking tour, visit the internationally renowned Robert C. Williams Museum of Papermaking, enjoy public art and performances at the Ferst Center for the Arts, or see iconic campus landmarks such as Tech Tower and the Kessler Campanile. Just beyond campus, Midtown offers a wide variety of restaurants and cultural attractions, including the High Museum of Art, Fox Theatre, Piedmont Park, and Atlanta Botanical Garden, while downtown destinations such as Centennial Olympic Park, the Georgia Aquarium, and the World of Coca-Cola are only a short trip away. Additional information about visiting Georgia Tech and exploring Atlanta is available at <https://www.gatech.edu/about/visit> and <https://discoveratlanta.com/>.

CONFERENCE DINNER:

The conference dinner will take place at Paschal's Restaurant, located at 180 Northside Drive SW, Atlanta, GA 30313 from 6pm to 9pm on October 13.

PARKING:

We recommend using the public parking facilities at the Student Center Parking Deck, located at 391 Ferst Drive NW, Atlanta, GA, 30332.

DAY 1: 12 OCTOBER 2026

Time	Room	Workshop
10:00-1:00	ASDL Cove	Hands-On with Rumoca: Modelica Source, Simulation, and Algebraic Backends in VS Code
10:00-1:00	ASDL Code	Physics + AI in Practice: Accelerating Modelica Workflows with Modelon Impact
10:00-1:00	SSC	SMArtInt+ - Hands-on AI: Generation and Integration of Neural Networks and Surrogate Models into Modelica
1:00-2:30	—	Lunch
2:30-5:30	ASDL Cove	Agentic Modeling and Simulation with Julia/Dyad
2:30-5:30	ASDL Code	Modeling HVAC and Control Systems with the Modelica Buildings Library
2:30-5:30	SSC	Introduction to Object Oriented Modeling, Simulation, Debugging and Dynamic Optimization with Modelica using OpenModelica

DAY 2: 13 OCTOBER 2026

Time	Session
8:00-8:30	Arrival
8:30-8:45	Conference welcome
8:45-9:50	Vendor sessions
9:50-10:05	Break
10:05-11:05	Vendor sessions
11:05-12:00	Keynote 1: Oliver Lenord
12:00-1:00	Lunch
1:00-2:40	Paper Session 1: FMI / Building Systems
2:45-3:15	Coffee break
3:15-4:30	Paper Session 2: Emerging Tools 1 / Power Systems
4:30-6:00	Transit to Paschal's

Schedule provided in Eastern Daylight Time (UTC-4)

DAY 3: 14 OCTOBER 2026

Time	Session
8:15–8:50	MA/FMI updates
8:50–9:45	Keynote 2: Daniel Mikkelsen
9:45–10:15	Coffee break
10:20–12:00	Paper Session 3: Emerging Tools 2 / Automotive
12:00–1:00	Lunch
1:00–2:15	Paper Session 4: Data Centers / Novel Applications
2:20–3:40	Industrial Presentations 1 and 2
3:45–4:15	Coffee break
4:15–5:00	Podium discussion

Schedule provided in Eastern Daylight Time (UTC-4)

DAY 1

10:00am - 1:00pm

ASDL CoVE

Hands-On with Rumoca: Modelica Source, Simulation, and Algebraic Backends in VS Code

James Goppert and Micah Condie (Purdue University)

ABSTRACT: This two-hour hands-on tutorial introduces Rumoca, an open-source, Rust-native Modelica compiler, and its VS Code development environment. Participants will follow a guided workflow from Modelica source through editing, diagnostics, compilation, simulation, visualization, and export to algebraic and numerical backends. Using curated example models, attendees will edit code, inspect diagnostics, run simulations, view results in the editor, and, where applicable, interact with an editable 3D view. From the same model, they will generate backend artifacts for CasADi, SymPy, JAX, and FMI-oriented workflows, then load representative outputs into Python or Jupyter to inspect equations and evaluate generated functions. The tutorial demonstrates a modern Modelica workflow in which one model serves as a reusable algebraic specification for simulation, optimization, automatic differentiation, machine learning, and model exchange. No prior Rumoca experience is required; basic Modelica, VS Code, and Python familiarity is helpful.

EXPECTED EXPERIENCE: No prior Rumoca experience is required; basic Modelica, VS Code, and Python familiarity is helpful.

SOFTWARE REQUIREMENTS: Installation of VS Code, Python

FURTHER INFORMATION: <https://rumoca.cognipilot.org/>

ASDL CoDE

Physics + AI in Practice: Accelerating Modelica Workflows with Modelon Impact

Mathias Strandberg and Pratyush Agarwal (Modelon)

ABSTRACT: AI assistants are becoming common in engineering software, but their value depends on whether they can support the physics-based work engineers actually need to do. In this hands-on workshop, participants will explore how AI-assisted development in Modelon Impact can help reduce workflow friction while keeping engineers in control of the modeling, simulation, and validation process. Using Modelon Impact, we will build and analyze an industry-relevant system model, progressing from component selection and model assembly through simulation setup, troubleshooting, and interpretation of results. Along the way, participants will see how the AI assistant supports common engineering tasks: surfacing relevant documentation, guiding model development, helping configure experiments, diagnosing errors and logs, and making results easier to understand. The session will also provide a practical introduction to Modelon Impact's cloud-based workflows and validated, robust Modelica library content. Participants will leave with a clearer understanding of what "Physics + AI" means in practice: AI that accelerates modeling work while preserving the transparency, reusability, and trust expected from physics-based simulation.

EXPECTED EXPERIENCE: Introductory-level knowledge of any Modelica

tool

SOFTWARE REQUIREMENT: Participants will be given temporary access to Modelon Impact

SSC

SMARTInt+ - Hands-on AI: Generation and Integration of Neural Networks and Surrogate Models into Modelica

Tim Hanke (XRG Simulation GmbH)

ABSTRACT: There is an increasing need for hybrid models consisting of physical and machine-learning or surrogate model parts. XRG has developed the SMARTInt+ Library, that enables the easy generation and integration of neural networks from different sources and types. This tutorial provides a hands-on introduction to the use of this library, which also includes a Python toolbox. The main use case is the creation and integration of a transient surrogate model for a thermodynamic application. This tutorial focuses on the generation and utilization of training data, as well as the definition and training of neural network architectures, with the objective of developing high-performance surrogate models. Participants will be introduced to different concepts and approaches for creating data-driven surrogate models and will gain an understanding of the underlying methodologies and best practices. A central part of the tutorial is the integration of the trained neural networks and surrogate models into Modelica models using the SMARTInt+ Interface-Blocks, allowing seamless substitution of physical components with learned surrogates. Additionally, several concepts of hybrid modeling that combine physical and data-driven approaches are introduced, with a strong focus on the practical integration of trained neural networks and surrogate models within the Modelica environment. Finally, model performance is evaluated through comparisons with the original physical model.

EXPECTED EXPERIENCE: Basic knowledge of neural networks/ AI and Python, as well as basic Modelica knowledge

SOFTWARE REQUIREMENT: Windows 10 or 11, Dymola 2026 Refresh 1 or OpenModelica 1.26 as a backup, and a Python IDE (PyCharm preferred)

FURTHER INFORMATION:

<https://xrg-simulation.com/en/seiten/smartint>

2:30pm - 5:30pm

ASDL CoVE

Agentic Modeling and Simulation with Julia/Dyad

Michael Tiller (JuliaHub)

ABSTRACT: In this workshop, we will present the latest information about Dyad, our platform for modeling and simulation. We will leverage the Dyad agent so participants can hit the ground running and explore all that the Dyad platform has to offer. Included in the presentation will be how to build graphical models from within VS Code. Discussions about best practices for using Dyad. Demonstrating how the Dyad agent flattens the learning curve and how it extends not just to modeling and simulation but assisting users with fully leveraging the incredible Julia ecosystem as well. Participants should come prepared with some tasks to give the agent so they can see how much progress they can make in a short amount of time.

EXPECTED EXPERIENCE: Beginner

SOFTWARE REQUIREMENT: Installation instructions for VS Code will be sent out in advance.

ASDL CoDE

Modeling HVAC and Control Systems with the Modelica Buildings Library

Michael Wetter, David Blum, Ettore Zanetti (LBNL)

ABSTRACT: This workshop introduces modeling for HVAC and control systems for buildings, districts, and data centers using the Modelica Buildings Library. While focused on these applications, the course is structured to benefit modelers of any thermo-fluid system. Following an overview of the library's scope, we will cover best practices for model configuration and troubleshooting. Through hands-on exercises, participants will develop simple HVAC system models, connect them to thermal loads, and implement feedback control. The instructors will demonstrate using OpenModelica, though participants are welcome to use Modelon Impact or Dassault Systemes' Dymola.

EXPECTED EXPERIENCE: Basic knowledge of Modelica. Having looked at tutorials of their Modelica modeling environment (OpenModelica, Modelon Impact, Dymola).

SOFTWARE REQUIREMENT: Recent version of OpenModelica, Modelon Impact or Dymola installed.

ADDITIONAL INFORMATION: <https://simulationresearch.lbl.gov/modelica/>

SSC

Introduction to Object Oriented Modeling, Simulation, Debugging and Dynamic Optimization with Modelica using OpenModelica

Adrian Pop (Linköping University), Luigi Vanfretti (Rensselaer Polytechnic Institute)

ABSTRACT: This tutorial gives an introduction to the Modelica language, the OpenModelica environment, and an overview of modeling and simulation in a number of application areas. Some advanced features of OpenModelica will be presented, including debugging, profiling, clocked synchronous support, real-time embedded code generation, OMPython, OMJulia, FMI export, etc. A number of hands-on exercises will be done during the tutorial, both graphical modeling using the Modelica Standard Library (MSL) and textual modeling. Bring your laptop for exercises.

EXPECTED EXPERIENCE: None

SOFTWARE REQUIREMENT: Install OpenModelica from openmodelica.org

ADDITIONAL INFORMATION: <https://www.ida.liu.se/~adrpo33/ModelicaTutorial2026/>

DAY 2

PAPER SESSION 1A: FMI

ROOM: ASDL CoVE 1:00PM - 2:40PM

Towards an FMI Layered Standard for DAE—Applications for Simulation and Optimization

Elmir Nahodovic, Andreas Heuermann, Joel Andersson, Adwait Verulkar, Srikanth Sivaramakrishnan, Masoud Najafi, Linus Langenkamp, Christian Bertsch, Erik Henningsson, Hans Olsson, Bernhard Bachmann

From Gazebo to FMI: A Modelica-Based Cyber-Physical Simulation Architecture for Firmware-Integrated Robotic Vehicles

Minhyun Cho, Minjoon Lee, Micah Condie, Michael Rooney, Evan English, James Goppert, Arslan Khan, Dongyan Xu, Inseok Hwang

Implementation of a High Fidelity Model of a Racecar into a Path Optimization Problem using Functional Mock-Up Units

Angel Zarate Villazon, Michael Balchanos, Dimitri Mavris

Terminals in SSP Dag Brück

PAPER SESSION 1B: BUILDING SYSTEMS

ROOM: ASDL CoDE 1:00PM - 2:40PM

Modelica Buildings Requirements Library

David Blum, Ettore Zanetti, Michael Wetter

Towards Replicable Implementations of ASHRAE Guideline 36 for Waterside Economizers

Julia Ho, Cary Faulkner, Wangda Zuo, National Laboratory of the Rockies

A Unified Multi-Fidelity Modelica Library for Desiccant Wheel Simulation

Hang Guan, Mingzhe Liu, Zhiyao Yang, Zheng O'Neill

A Templated Modelica Model for Decoupling Physical System and Control in Resilient Community Energy Systems

Juan Diego Flores Garcia, Tahiya Hossain, Viswanathan Ganesh, Wangda Zuo

PAPER SESSION 2A: EMERGING TOOLS 1

ROOM: ASDL CoVE 3:15PM - 4:30PM

Rumoca: Modelica as a Universal Algebraic Frontend via a Rust-Native Compiler

Micah Condie, Abigaile Woodbury, Thomas Meschede, Joel Andersson, James Goppert

Differentiable Parameter Optimization for DAEs with State-Dependent Events

Ion Matei, Maksym Zhenirovsky, Anthony Wong

ModelScript: Rethinking the Digital Thread with a Web-Native, Polyglot Compiler for Modelica, SysMLv2 & Beyond

Mohamad Omar Nachawati

PAPER SESSION 2B: POWER SYSTEMS**ROOM: ASDL CoDE 3:15PM - 4:30PM****MMChvdcLibrary: Modeling Arm Average Value Modular Multilevel Converters in Modelica for Power System Simulation**

Md Nazmul Hassan Chowdhury, Luigi Vanfretti

Real-time simulation and CHIL testing of power system stabilizers on microcontrollers with Modelica and eFMI

Luigi Vanfretti, Christoff Bürger, Joseph Pizzimenti, Kyle Wilt, Hao Chang

Deploying OpenIPSL Models into DigSILENT PowerFactory via FMI Model Exchange

Hao Chang, Luigi Vanfretti

DAY 3

PAPER SESSION 3A: EMERGING TOOLS 2**ROOM: ASDL CoVE 10:20AM - 12:00PM****Discontinuity-Aware Solvers for Vapor-Compression Cycles**

Shreyas Ekanathan, Avinash Subramanian, Oscar Smith, Yingbo Ma, Chris Rackauckas, Christopher Laughman

Toward Equation-Based Building Energy System Modeling in Julia: A Modular Prototype and Validation Study

Ruizhi Song, Mingzhe Liu, Zhiyao Yang, Zheng O'Neill

Implementation of OpenIPSL in Dyad

John Batteh, Luigi Vanfretti

PAPER SESSION 3B: AUTOMOTIVE**ROOM: ASDL CoDE 10:20AM - 12:00PM****A Single-Element Multi-Point Compliance for Rigid-Body Connections**

Behzad Danaei, Srikanth Sivaramakrishnan

A 2-DOF MultiBody Track Joint for Race Car Optimization

Srikanth Sivaramakrishnan, Adwait Verulkar, Behzad Danaei, Ryan Trinter

A Singularity Avoiding Track Model for Race Car Optimization

Adwait Verulkar, Behzad Danaei, Srikanth Sivaramakrishnan, Ryan Trinter

Five-Link Rear Suspension for F1: A Two-Stage Agentic AI Study with the Vehicle Dynamics

Library Rebecka Eldh, Johan Andreasson, Mathias Strandberg

DAY 3 SCHEDULE

PAPER SESSION 4A: DATA CENTERS

ROOM: ASDL CoVE 1:00PM - 2:15PM

A Modelica-based Virtual Testbed for Hybrid-Cooled Data Centers

Hongjun Li, Viswanathan Ganesh, Wangda Zuo

Integrated Microclimate Model in Modelica for Data Center Chiller Farms

Kaustubh Phalak, Pouya Javidmand, Zhanwei He, Gang Wang, Sean Guo

Agent-Assisted Construction of Modelica System Models from Validated Components: A Data Center Cooling Case Study

Nithish Selvan, Johan Andreasson, Pratyush Agarwal

PAPER SESSION 4B: NOVEL APPLICATIONS

ROOM: ASDL CoDE 1:00PM - 2:15PM

A Unified Dynamic System Testing Framework

Scott Greenwood

Data-Driven Refinement of a Native Modelica Model Predictive Controller for Multi-Variable Gas Turbine Hydrogen Blending Control

James Harper, Paolo Pezzini

Quantum Algorithms for Patent Portfolio Analysis Using OpenModelica and IBM Qiskit

Mark Myers

INDUSTRIAL PRESENTATIONS 1

ROOM: ASDL CoVE 2:20PM - 3:40PM

A Novel Modular Framework for Unified Data Center Cooling Virtual Testbed

Viswanathan Ganesh, Hongjun Li, Michael Maloney, Wangda Zuo

Rapid De-Risking and Sizing of GSHP Systems in Modelica

Victor Braciszewski, Lone Meertens, Lieve Helsen

De-Risking Energy System Integration through Platform-Based Design and Modelica

Michael Wetter, Matthias Sulzer

A Workflow for Enabling Modelica Building and District Models for DRTS-Ready FMUs

Nicholas Long, Brian Ball, Andy Olson, Nischal Guruwacharya, Rebecca Barney, Whitney Trainor Guitton

INDUSTRIAL PRESENTATIONS 2

ROOM: ASDL CoDE 2:20PM - 3:40PM

A Free-Open-Source Modelica Simulation Package for Air Conditioning and Refrigeration Cycle (SPARC)

Chenjiyu Liang, Donghun Kim, Jiacheng Ma

Real-Code Models for Electromagnetic Transient Simulations: Interoperability with Modelica

Luigi Vanfretti, Hao Chang

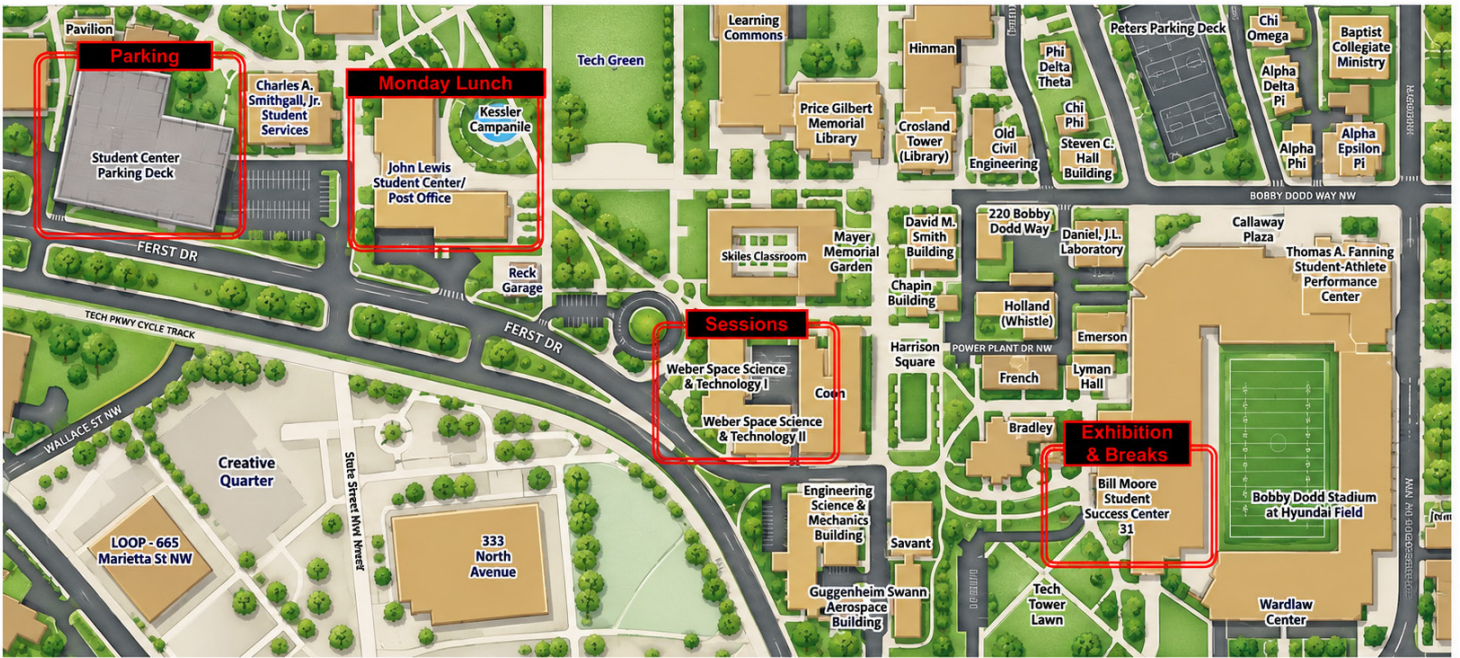
Optimization of a Green Hydrogen Production Case Study Utilizing a Sigmoid Function

Nicholas Pilot, Garrett Doty, James Harper, Behrooz Motealleh, Paolo Pezzini

Simulating Renewable Load-Following Hydrogen Electrolysis Plant Controls

Garrett Doty, Nicholas Pilot, James Harper, Paolo Pezzini

Campus Overview



ASDL Building



KEYNOTE SPEAKERS





Oliver Lenord

Research and Development Engineer, Leader of OpenScaling project
Bosch Corporate Research

BIO

Oliver Lenord earned his Phd at the Mechatronics Institute of the Univ. of Duisburg, Germany. Since 2016 he is affiliated with Robert Bosch - Corporate Research. Formerly he led the simulation software development group at Bosch Rexroth and worked as product manager for Siemens PLM in California, USA. Currently he is leading the European ITEA4 project OpenSCALING (openscaling.org). Earlier he also initiated and led the German research project PHyMoS, concerned with integrating scientific machine learning with classical modeling and led the award winning ITEA3 project EMPHYSIS that delivered the new eFMI standard. He also serves as vice-chairman of the Open Source Modelica Consortium (OSMC).

ABSTRACT

Pushing the Limits by Open Standards: Enabling Larger, Faster, and More Credible System Simulations

The scalability of virtual engineering has become an increasingly critical demand driven by the innovation pressure towards carbon neutral, energy-efficient and cost-effective solutions. Open standards like Modelica, FMI, eFMI and SSP play a major role in making system simulation accessible to a broad user base, fostering a rich ecosystem of research and technology development across academia and industry. The European publicly funded project OpenSCALING (Open Standards for SCALable Virtual EngineeriNG) has further strengthened this ecosystem through standard enhancements and new methods, enabling simulation and credible processes to scale to the next level. This talk will highlight key project achievements in the fields of:

- Modelica compiler enhancements
- Integration of scientific machine learning with Modelica
- Uncertainty quantification and credible workflows

Reference to newly developed methods and standard enhancements are given including array-preserving compilation, pre-compiled components, and improved support of sensitivities in FMI. In addition the newly proposed layered standards Is-sa (for sensitivity analysis) and Is-uq (for uncertainty quantification) are briefly introduced.

Benchmarks and examples will illustrate these advances, demonstrating the ability to:

- Increase maximum model size by orders of magnitude
- Drastically reduce compile and simulation start-up times
- Accelerate simulations by orders of magnitude using surrogate modeling and machine learning
- Integrate and efficiently process Artificial Neural Networks within Modelica
- Leverage FMI in diverse machine learning environments for robust model training
- Enable traceable credibility workflows through standardized metadata in FMI and SSP
- The practical relevance of these advances will be demonstrated referring to selected industrial use cases from different application domains.

Finally, an outlook will be given on the future potential of agentic AI for Modelica and the proposed Base Modelica language.



Daniel Mikkelson

Integrated Energy Modeling and Simulation Engineer

Affiliation: Idaho National Laboratory

BIO

Dr. Daniel Mikkelson is an integrated energy modeling and simulation engineer at Idaho National Laboratory. His research interests include systems modeling and analysis of advanced nuclear reactor systems, thermal energy storage, experimental systems, and integrated energy systems leveraging nuclear power in electric and non-electric applications. Analyses have investigated control strategy, capacity optimization, dispatch scheduling, and evaluation of integrated system feedback. Dr. Mikkelson is the lead developer of HYBRID, the Modelica modeling repository of the Integrated Energy Systems program at INL. Dr. Mikkelson received his Ph.D. in Nuclear Engineering from North Carolina State University in 2021.

ABSTRACT

Nuclear System Modeling for Integrated Energy Systems

Analyses

The HYBRID modeling repository has been developed to analyze the control, dispatch, and feedback of integrated energy systems that leverage nuclear power for thermal and electrical applications. As global priorities endorse nuclear energy, enabling rapid, open-source, dynamic evaluation of potential nuclear applications can enable faster development of integrated systems that will improve the utilization of nuclear energy in diverse applications for both brownfield and greenfield integrations. An overview of the nuclear approach to modeling and simulation will present areas where Modelica presents an ideal tool to investigate configuration viability in dynamic deployment scenarios.

ABSTRACTS



Towards an FMI Layered Standard for DAE

Applications for Simulation and Optimization

Elmir Nahodovic ¹ Andreas Heuermann ² Joel A. E. Andersson ³ Adwait Verulkar ⁴
 Srikanth Sivaramakrishnan ⁴ Masoud Najafi ⁵ Linus Langenkamp ⁶ Christian Bertsch ⁷
 Torsten Sommer ¹ Erik Henningsson ¹ Hans Olsson ¹ Bernhard Bachmann ⁶

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Abstract

The Functional Mock-up Interface (FMI) 3.0 standard for Model Exchange is restricted to hybrid ordinary differential equations, requiring internal algebraic equations to be solved inside the Functional Mock-up Unit (FMU) before derivatives are returned to the importer. For models originating from, e.g. Modelica, this typically introduces internal Newton iterations for non-linear algebraic systems. These nested solves hide the algebraic structure from the importer, can reduce accuracy and increase computational cost, introduce hidden solver states, and may cause robustness problems in simulation and gradient-based dynamic optimization.

In this article, we present *fmi-ls-dae*¹, a proposal for an FMI layered standard that exposes algebraic equations and their associated algebraic variables as part of a semi-explicit index-1 differential-algebraic equation. During Continuous-Time Mode, the importer obtains state derivatives, algebraic residuals, and outputs as functions of the continuous states, algebraic variables, inputs, and time, and is responsible for satisfying the residual equations. The proposal adds a layered-standard XML manifest describing algebraic variables, residuals, and their model structure and dependencies. No extension of the FMI C API is required, and the layered-standard mechanism is designed so that tools without DAE support can ignore the additional information.

The proposal is demonstrated end-to-end through prototype implementations in several independent tools. Dymola and CasADi generate DAE-FMUs, while CasADi, FMIOPT, Simcenter Twin Activate, fmusim, and MOO, the dynamic optimization tool of OpenModelica, import them for simulation and optimization. In Simcenter Twin Activate, `SimpleDAE.fmu` matches an independently

implemented DAE model and `Fourbar1.fmu` (from the Fourbar1 model in the Modelica Standard Library) gives results identical to the corresponding Dymola and OpenModelica simulations. The Robertson Reference FMU² is simulated in fmusim with IDA, while a CasADi example solves a direct-collocation optimal control problem that takes the load of a crane, exported as a DAE-FMU, to a desired position under a quadratic cost.

The main optimization example considers an industrially relevant multilink rear suspension corner with a non-linear kinematic loop. The same model is exported by Dymola as an *fmi-ls-dae* DAE-FMU and as a standard FMI 3.0 ODE-FMU. FMIOPT transcribes the trajectory-tracking problem on 150 intervals using three-point Radau IIA collocation. With the DAE-FMU, the algebraic variables are optimization variables and the exposed residuals are imposed directly as equality constraints; IPOPT converges cleanly and follows the reference essentially exactly. With the equivalent ODE-FMU, where the kinematic loop is closed by an FMU-internal Newton iteration, IPOPT fails with *Restoration Failed!* after 369 iterations and produces unphysical solution artifacts. The DAE-FMU formulation is independently solved with MOO using 400 discretization points and converges in 57 NLP iterations, with the actuator force agreeing with the FMIOPT solution. Consistent initialization, event handling, and higher-index formulations remain under active development.

Keywords: *FMI, Modelica, DAE, Differential Algebraic Equations, Layered Standard, Model Exchange, Dynamic Optimization*

¹GitHub: [modelica/fmi-ls-dae](https://github.com/modelica/fmi-ls-dae)

²GitHub repository: <https://github.com/modelica/Reference-FMUs>

Firmware-in-the-Loop Testing of Robotic Vehicles

Minhyun Cho¹ Minjoon Lee¹ Micah K. Condie¹ Michael P. Rooney² Evan R. English² James M. Goppert¹ Arslan Khan³ Dongyan Xu² Inseok Hwang¹

¹School of Aeronautics and Astronautics, Purdue University, United States

²Department of Computer Science, Purdue University, United States

³School of Electrical Engineering and Computer Science, The Pennsylvania State University, United States

Abstract

Firmware-level cyber-physical testing of robotic vehicles often requires the same embedded firmware to be evaluated across different physical models, fidelities, disturbances, and sensor or actuator fault conditions. We present a Modelica/FMI-based architecture that places a replaceable FMI 3.0 Co-Simulation FMU behind a stable firmware interface. Vehicle dynamics, actuators, sensors, and environmental effects are modeled in Modelica and exported through Rumoca. Our physics backend, COURBET, hosts the FMU and couples it to unmodified ArduPilot firmware binary running in FastDyn's QEMU-based firmware-in-the-loop environment. The architecture is demonstrated on copter, rover, and fixed-wing configurations and is designed to support controlled model substitution, fidelity variation, fault injection, and scalable formal-methods workflows.

Keywords: *Modelica, FMI 3.0, Functional Mock-up Unit (FMU), Firmware-in-the-Loop (FITL), Cyber-Physical Systems, Robotic Vehicle Simulation*

1 Motivation and Architecture

Robotic simulators provide mature physics, rendering, and sensor environments, but simulation-level experiments often become tightly coupled to simulator-specific plugins, model descriptions, and timing semantics. This coupling is inconvenient when the physical plant must be systematically varied while preserving the same execution environment. For example, Gazebo provides dedicated systems for common vehicle components such as multi-copter motor-rotor models, whereas introducing a different propulsion model, such as a jet engine, generally requires simulator-specific model or plugin development.

We address this limitation by placing a replaceable physical-model architecture between the firmware execution environment and the plant. As illustrated in Fig. 1, FastDyn executes an unmodified ArduPilot firmware binary in QEMU using instruction-counted virtual board time, while COURBET provides the interface to the physical backend. In the proposed architecture, the plant is described in Modelica and exported through Rumoca as an FMI 3.0 Co-Simulation FMU. At each board-timer

synchronization boundary, COURBET transfers actuator commands to the FMU, advances the plant FMI 3.0 interfaces, and returns updated physical and sensor states before firmware execution resumes. We keep the firmware-facing interface unchanged while allowing the physical model to be replaced independently. Vehicle dynamics, actuator models, sensor models, environmental effects, and fault parameters can therefore be varied without altering the simulator architecture.

2 Evaluation and Implications

We demonstrate the architecture with ArduCopter, ArduRover, and ArduPlane configurations using the same FMI-centric execution workflow. In 60 s virtual-time ArduCopter experiments, the FMI backend achieved an effective backend real-time factor of $299.6\times$, approximately 34.8 times higher than the $8.6\times$ measured at the legacy asynchronous Gazebo coupling interface, while instrumented FMU advancement accounted for only $1.508 \pm 0.065\%$ of total wall-clock execution time. The contribution is architectural rather than a claim that Modelica/FMI universally replaces robotics simulators. The proposed interface provides a controlled and reusable physical-model boundary for firmware-integrated experiments, supporting model substitution, fidelity variation, parameter sweeps, Monte-Carlo studies, fault injection, fuzzing, and falsification.

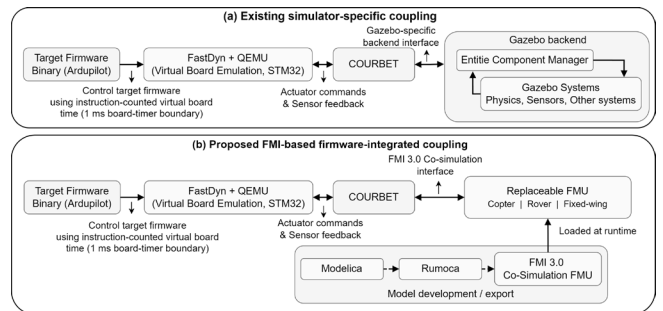


Figure 1. Comparison of (a) existing simulator-specific plant coupling and (b) the proposed FMI-based plant coupling for firmware-in-the-loop (FITL) execution. The simulator-specific backend interface is replaced by a standard FMI 3.0.

Implementation of a High Fidelity Model of a Racecar into a Path Optimization Problem using Functional Mock-Up Units

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Abstract

Racecar trajectory optimization relies on several approaches that span data-based techniques, reinforcement learning, and optimal control. Optimal control poses a significant challenge as it computes this optimal path based on the solution to detailed equations of motion. As such, there have been very few implementations that use a high-fidelity model such as those implemented for a driver-in-the-loop simulator, which are commonly written using the Modelica language. The main reason is that a number of simplifying assumptions are needed to make the problem compatible with optimal control solutions, threatening the validity of the solutions found as well as the consistency of the insights derived from it in the engineering process.

This paper investigates the feasibility of integrating an open-loop, high-fidelity Modelica vehicle model into an optimal path problem solved via direct collocation. The vehicle model, which was originally developed by an OEM for a driver-in-the-loop simulator in a top-tier motorsport series, is integrated through Model Order Reduction and the Functional Mock-up Interface (FMI) standard. Optimization is performed in Dymos using an OpenMDAO-based reduced-order model informed dynamically by the high-fidelity FMU. To address the computational challenges of this workflow, experimental tests evaluate both optimizer behavior and FMU instantiation, specifically state serialization, which enables discrete node evaluations across the collocation mesh from recorded baseline data. The conducted experiments and their results are summarized below:

M1 - FMU serialization comparison: This experiment tests whether a simulation in the open-loop driver can be reconstructed using serialization. The results show perfect agreement between the simulation stopped and restarted using serialization and the one completed in a single pass. Similarly, it presents good agreement with the original data recorded from the Driver-in-the-Loop simulator.

M2 - FMU sensitivity to changes: This experiment tests whether the FMU remains sensitive to changes in the controls after being started from a serialization file. Additionally, it tests the timestep necessary to see any changes for this particular implementation. The results indicate controllability of the model after being serialized,

and identify a timestep of 6 milliseconds to see noticeable change.

M3 - Connection between FMU and lower-fidelity model: This experiment tests the connection of vehicle states between the FMU and the lower-fidelity model. Some changes were observed after executing the optimization problem for one iteration, which suggests a correct connection of the FMU as it is enforcing different dynamics that are informed by the FMU.

O1 - Compatibility with external models: This experiment centers on determining the feasibility of the optimizer to interface with external models. It was demonstrated that an externally compiled tire model can be implemented and yield convergent results, confirming the potential of external models in Dymos.

O2 - Node location: As the proposed approach requires the precise location of the discretization nodes to create serializations of the FMU based on the initial guess data, an experiment to define the node location is conducted. The results show the distribution is not equidistant in the Cartesian coordinates, motivating possible future investigation based on the path's curvature.

O3 - Derivative Strategy: This experiment defines a strategy to evaluate the role of derivative precision by evaluating the progression of the states to identify signs of divergence. It was tested using finite differences and it was observed that possible divergences may exist, with further testing limited due to unresolved license issues.

These results establish FMU state serialization as an architectural mechanism for incorporating dynamic vehicle models into direct collocation problems. Serialization allows the high-fidelity model to be initialized deterministically at individual nodes using baseline telemetry aligning closely with collocation's discretization approach. While finite-difference derivative approximations introduced numerical instability that hindered end-to-end convergence, the state coupling and serialization pipelines were verified. Future work will center on resolving the gradient calculation bottleneck through algorithmic differentiation, adjoint sensitivity formulations, and curvature-adaptive node distribution schemes to achieve full convergence with production-grade vehicle models.

Terminals in SSP

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1 Introduction

System Structure and Parameterization (SSP) is a tool independent standard to define complete systems. In a typical use of SSP, the system description consists of several interconnected Functional Mockup Units (FMUs), nested system descriptions, and their parameterizations and other related data, as shown in Figure 1.

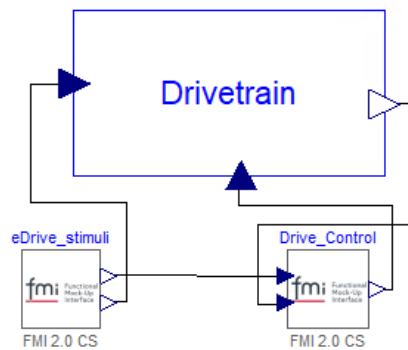


Figure 1. Simple hierarchical system model.

Structured connectors (ports) are available in many simulation languages, including FMI (Functional Mockup Interface) and Modelica. However, such a concept is missing in SSP making it incompatible with its closest related standards.

2 Why Terminals?

Terminals offer several system design advantages.

- Grouping of connectors (input, output, acausal, nested terminals) so they can be managed as one unit, for example to make a connection. Hence the maze of connection lines is reduced.
- Separation of terminal definition from its use, especially that the terminal can be changed without affecting the connection topology.
- Decluttering of the namespace, as members of the terminal are local.

3 Requirements on Terminals

SSP may require a terminal concept that comes in two flavors: one to provide FMI 3 compatibility, and another more general end better encapsulated for SSP systems.

A concrete list of requirements include:

- Support for FMI 3 terminals with externally defined connectors.
- Support for encapsulated terminals where connectors are defined within the terminal.
- Possibility to define common terminal types, which can be used to express common interfaces for many components and systems.
- Optionally reuse the terminal concept to define parameter records (as parameters are a particular kind of connector in SSP).
- Optionally to define parameter sets (SSV) with values of record type.

4 Design Proposal

FMI terminals have, for compatibility reasons, certain properties that do not lend themselves well to SSP. The key conceptual differences are

- In FMI, the connectors are defined individually, and the optional terminal groups them.
- In SSP, the terminal defines the connectors, but they can be used to define individual connectors with the same name if needed.

A unification is possible if we stretch the SSP semantics and require that FMI terminals are used for connections if they have been defined.

An example demonstrates the design.

```
<ssd:Component name="Source_fmu">
  <ssd:Connectors>
    <ssd:Connector kind="undefined"
      name="tSource">
      <ssd:Terminal name="tSourceType">
        <ssd:TerminalMapping
          memberName="pos"
          variableName="tSource_pos"/>
        <ssd:TerminalMapping
          memberName="neg"
          variableName="tSource_neg"/>
        <ssd:TerminalMapping
          memberName="feedback"
          variableName="tSource_feedback"/>
      </ssd:Terminal>
    </ssd:Connector>
    <ssd:ElementGeometry . . . />
  </ssd:Component>
```

The Modelica Buildings Requirements Library for In-Simulation Requirements Verification for HVAC Applications

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Abstract

Building and district energy systems and their controls are facing new performance objectives with stricter reliability criteria, such as providing grid flexibility, critical cooling to data centers, multi-sector coupling of district networks, and lower energy costs for building owners. Verifying that operational requirements are met throughout the system design process using dynamic simulations leads to lower commissioning costs, higher up-time, less failure risk, and provides a basis for system specification and contractual agreements. However, manually verifying operational requirements from simulation results is a lengthy, tedious, and repetitive task for modelers, especially for large and complex systems. The open-source Modelica Requirements Library was previously developed for defining requirements within a Modelica model and verifying them during a simulation.

This paper presents the open-source Modelica Buildings_Requirements Library, which extends the Modelica Requirements Library with template blocks representing common requirements for building and district energy and HVAC system applications. The paper describes the library's structure and component models and demonstrates its use to verify the operational performance of a novel commercial building HVAC central plant.

The Modelica Buildings_Requirements Library has four component models and two base class models at this time. The component model `GreaterEqual` verifies whether an input signal is greater or equal to a second input signal when the test is active, with optional delay time after activation and with optional minimum time of violation to be considered violated. The component model `MinimumDuration` verifies whether a signal is true for a minimum time. The component model `StableContinuousSignal` verifies whether an input signal is stable when the test is active, with optional delay time after activation and with optional minimum time of violation to be considered violated, checking that the signal does not have a high amount of oscillations. Finally, the component model `WithinBand` verifies whether an input signal is within constant lower and upper bounds when the test is active, with optional delay time after activation and with optional minimum time of violation to be considered violated. A validation model shows example

performance of the `GreaterEqual` model under different conditions, demonstrating when the block reports that a requirement is satisfied, violated, or untested.

The Modelica Buildings_Requirements Library is demonstrated using a simulation of a novel, complex commercial building central plant design known as the Time-Independent Energy Recovery (TIER) plant, as implemented in the Modelica Buildings Library v12.0.0 as `Buildings.DHC.Plants.Combined.AllElectricCWStorage`. TIER integrates heat recovery chillers, cooling-only chillers, a thermal energy storage tank, cooling towers, and a "trim" heat source, such as air source heat pumps. Example results of using the Modelica Buildings_Requirements Library in this case show that 80 requirements were defined and 15 of them were reported to be violated during the simulation. A closer look at the reported violations of the requirement to maintain the plant-wide chilled water supply temperature within 0.5K of the set point shows that the requirement is satisfied most of the time, or undecided due to chilled water production being inactive. However, one violation occurs the first time chilled water is called for, where it takes extra time for the volume of water in the chilled water loop to cool down to the set point from an initial condition, and a second violation occurs when the cooling load on the plant reaches its peak, when there is about a 2 hour window when the chilled water temperature set point is not maintained, with a peak excursion of about 2K.

In conclusion, the Modelica Buildings_Requirements Library proved useful to define and observe operational violations that quickly led to identification of control issues to be addressed in the example project, albeit at a price of some increased simulation time. There are considerations for future work. While the library indicates when requirements are violated, it does not indicate the severity of violations, such as how long, how often, nor by how much a violation occurs. This information would be useful to aid users in prioritizing violations to address. Furthermore, being able to formally define acceptance criteria would allow for short, small violations and give users an even more accurate assessment of desired performance. Lastly, developing and distributing more equipment and system-specific requirements templates would make it easier for users to specify and use requirements in their own model-based workflows.

A Workflow for Transparent and Replicable Modelica Implementations of ASHRAE Guideline 36 Controls for Chiller Plants with Waterside Economizers

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Keywords: rule-based controls, chiller plant, controls, ASHRAE Guideline 36, waterside economizer, modelica

1 Introduction

Control implementation in Modelica can be arduous, and it has been under-leveraged for G36 control analysis in chiller plants. This work expands on previous research and the authors' experience implementing complex rule-based controls in Modelica by defining and demonstrating a replicable method to implement complex rule-based controls consistent with sequences such as G36 in Modelica. Using this workflow, the authors develop fourteen control blocks and fifty control actions related to G36 sequences for a case study of a chiller plant with a waterside economizer in Modelica. Examples from the case study demonstrate the full scope of the workflow, from documenting written controls to troubleshooting full model control instability, with the goal of making the workflow understandable and replicable by future researchers.

2 Methodology

The control development workflow is described in detail using the examples from the case study, and a modeling testbed of a chiller plant with a WSE from (Fan et al. 2021) is used to evaluate G36-consistent controls. Fourteen control blocks describing around 50 control actions implemented using the workflow are described. The main steps of the workflow are shown below.

1. G36 is thoroughly reviewed, and a centralized tracking document (CTD) is created.
2. Control blocks are developed for subgroups of control actions within each control sequence.
3. Each control block is subject to open loop testing for each control action it executes, and associated testing information is stored in the CTD.
4. Once all control blocks pass open loop testing, they are integrated into subsystem controllers and then into a partial control block.

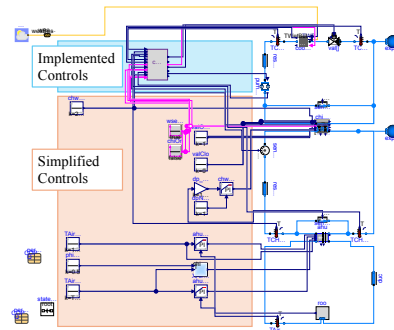


Figure 1. A full system model with around half of the implemented control blocks replaced with simple control signals.

5. The partial control block is integrated with a physical system model and is subject to closed loop testing. If control issues are found, the workflow is re-iterated until the model behaves as expected.
6. If unexpected results are not easily attributable to a control block, the controls are systematically simplified to identify and correct at-fault control blocks (Figure 1).

3 Preliminary Results

Results from open and closed loop testing are presented to demonstrate the control development workflow. An example of closed loop testing is a control instability that propagates through the entire model and therefore makes identifying the at-fault control block difficult. Using step 6 of the replicable control workflow described in this paper (Figure 1), the relevant control block is isolated, and the issue is diagnosed as unstable fan speed PI controls that are exacerbated by large numbers of state events, which are generated from overly complex implementation of fan enable controls.

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A Unified Multi-Fidelity Modelica Library for Desiccant Wheel Simulation

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Abstract

This work presents an open-source Modelica library for desiccant wheels that combines dynamic physics-based, effectiveness-based, and performance curve-based models behind a common four-port, plug-and-play interface, allowing model fidelity to be changed without restructuring the surrounding system. Its configurable full-wheel formulation additionally predicts the process- and regeneration-side outlet states simultaneously. Validation establishes the credibility of the physics-based implementation. Demonstration and runtime results establish the flexibility and intended application of each fidelity level.

Keywords: *Desiccant wheel; acausal modeling; physics-based models; reduced-order models; heat and mass transfer*

1 Motivation and Objective

Detailed desiccant-wheel models support component design and control development, whereas reduced-order models suit long-term system simulation. Existing implementations are commonly proprietary or lack reusable, connector-based interfaces. Moreover, conventional single-element formulations represent one alternating sector and do not directly provide simultaneous device-level outlet states. This work addresses these limitations through a unified Modelica library.

2 Modeling Framework

Built on the Modelica Buildings Library, it is organized into *PhysicsBased*, *ReducedOrder*, and *Examples* packages. All device models expose the same four-port fluid interface and are mutually replaceable.

The physics-based models solve one-dimensional coupled energy and moisture conservation equations for the air and desiccant phases. Configurable isotherms, geometry, sector allocation, and rotational speed support different designs and operating strategies. The single-element formulation discretizes a representative honeycomb channel axially and switches its mode and flow direction between sectors. The multi-element formulation assembles these elements circumferentially and advances their logical positions at discrete switching events. Elements occupy both sectors at once, and their outlet states are aggregated separately for the process and regeneration streams,

preserving cyclic sorption dynamics while providing full-wheel outlet states.

Two steady-state alternatives retain the same connectors for computationally demanding applications: an effectiveness-based model using characteristic thermodynamic potentials and a curve-based model using polynomial performance correlations.

3 Validation and Main Findings

The single-element physics-based model reproduces the reference transient profiles reported by Zhang et al. (2003) over a complete rotation cycle. Temperature prediction achieved an RMSE of 0.76 °C and an R^2 of 0.991. For the humidity ratio, the corresponding values were 0.0016 kg/kg_a and 0.972. The multi-element formulation was validated separately in prior work (Guan et al. 2026).

Six demonstration cases vary discretization, geometry, sector allocation, and material through parameter changes alone. In a seventh, time-varying regeneration temperature and wheel speed shift both the mean value and the oscillation amplitude of the outlet states. For 3600-s simulations in Dymola on an Intel Core i7-12700 workstation, the 6- and 12-element physics-based cases required 1.7 and 16.5 s, whereas the effectiveness- and curve-based models required about 3 ms, roughly three orders of magnitude faster. The reduced-order models therefore suit annual or optimization studies, although their empirical parameters remain application-dependent.

Significance and availability. The acausal, replaceable architecture supports component design, control development, and system assessment, and extends to total energy recovery wheels. The library is available under the MIT License at <https://github.com/BE-HVACR/DesiccantWheel>.

Acknowledgment. This work was partially supported by U.S. Department of Energy award DE-EE0010850.

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A Templated Modelica Model for Decoupling Physical System and Control in Resilient Community Energy Systems

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Abstract

As extreme weather events increasingly threaten the aging U.S. electrical grid, microgrids capable of islanded operation have become essential for sustaining community resilience during prolonged outages. While high-fidelity Modelica models can accurately capture microgrid dynamics, the manual effort required to build and reconfigure distinct system architectures for each equipment combination severely limits practical design exploration. Existing planning tools such as HOMER and REopt rely on simplified, static control logic and mixed-integer linear programming, which cannot resolve fast transient dynamics or support advanced supervisory control testing. Meanwhile, existing high-fidelity Modelica frameworks tightly couple control logic to physical equipment configurations, making it costly to swap topologies or control strategies, since any change to the equipment mix typically demands substantial manual reengineering of the graphical model layer.

This paper addresses these limitations by presenting a modular Modelica template framework for islanded-capable Net Zero Energy Community (NZEC) microgrids that structurally decouples the physical distributed energy resource (DER) system from its supervisory controller. A parameterized 1D array switching mechanism automatically instantiates the correct physical topology and corresponding baseline controller from among 12 distinct system configurations combining diesel generators (DG), battery energy storage (BESS), photovoltaics (PV), and wind turbines—without requiring manual model rewiring. Dedicated controller logic governs dispatch for DG-only, BESS-only, and combined DG–BESS strategies, each employing threshold-based state machines that respond to real-time load, generation, and state-of-charge signals communicated over standardized information buses. Both the physical system and controller are exported as Functional Mock-up Units (FMUs) and orchestrated via an external Python script that automatically assigns equipment capacities and executes co-simulations across the full design space, enabling rapid, repeatable batch evaluation without manual intervention.

The framework's utility is demonstrated through a case study of the Historic Green Village, a mixed-use commu-

nity on Anna Maria Island near Tampa, Florida, simulating a 168-hour (7-day) hurricane-induced outage with a peak critical demand of 31.5 kW and a weekly energy requirement of 3,669.4 kWh. All 12 templated topologies were independently instantiated and validated, confirming correct automatic assembly of equipment and control logic for every configuration, including those incorporating wind resources deemed unviable for this specific site. An exhaustive parametric search across PV and BESS sizing identified configurations capable of sustaining the full critical load, evaluated against capital cost, fuel consumption, and CO₂ emissions. Rather than isolating a single optimal design, the study mapped a Pareto-optimal frontier via a non-dominance filter, yielding seven viable configurations spanning trade-offs from lowest capital cost (DG-only, \$32K, 1,281 kg CO₂) to zero-emissions operation (BESS+PV, \$450K, 0 kg CO₂), with intermediate hybrid DG+BESS+PV configurations offering balanced and low-emissions alternatives. Time-series dispatch analysis of four representative architectures—DG-only, purely renewable PV+BESS, and two hybrid Pareto extremes—further validated the dynamic stability of the baseline control strategies under realistic load and weather conditions, revealing distinct operational signatures such as continuous diesel cycling in cost-optimized designs versus reduced generator activation and preserved battery health in emissions-optimized designs.

By decoupling physical and control models, this templating approach establishes a time-efficient, reproducible foundation for future microgrid design workflows—enabling researchers to rapidly implement, test, and benchmark novel supervisory control algorithms against established baseline strategies without rebuilding models from scratch. Because the physical plant is fully decoupled from the operational control logic, this framework provides a scalable basis for future work integrating advanced multi-objective optimization algorithms that jointly consider equipment sizing and dynamic dispatch strategy, ultimately supporting the design of more resilient, cost-effective, and low-carbon net-zero energy communities capable of withstanding an increasingly volatile climate.

Keywords: microgrid, resilience, template models, design

Rumoca: Modelica as a Universal Algebraic Frontend via a Rust-Native Compiler

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Abstract

Modelica is a mature language for declarative modeling of cyber-physical systems, but many modern engineering workflows (optimization, differentiable simulation, scientific machine learning, and symbolic analysis) live in CasADi, JAX, SymPy, and Julia. Existing Modelica toolchains do not target these environments directly, so the same model is often written several times with no guarantee that the versions agree. Rumoca is a Rust-native Modelica compiler that turns Modelica into a universal algebraic frontend: one declarative source, emitted into whatever ecosystem the work demands.

Keywords: *Modelica compiler, code generation, Rust, WebAssembly, FMI, CasADi, differentiable simulation*

The compiler is organized as a sequence of explicit phase boundaries, each producing an inspectable, typed intermediate representation: a Parsed IR, a Flattened IR with object-oriented structure and `connect` equations resolved, a DAE IR in the hybrid form of Modelica Language Specification Appendix B, and a Solver IR lowered to a register-based opcode stream with automatic-differentiation Jacobians. Downstream consumers read these IRs but never influence them, which is what lets one compiler serve many ecosystems rather than a simulator alone. Code generation is template-driven, and shipped backends include CasADi, SymPy, JAX, Julia/ModelingToolkit, ONNX, JavaScript, dual-mode FMI 2.0 and 3.0 FMUs, and an experimental embedded C target.

We will report what the compiler currently handles, measured against the 566 root example models of the Modelica Standard Library 4.1.0. Rumoca flattens 99 % of them, constructs a DAE for 73 %, and integrates 28 % to their experiment stop time, with roughly two-thirds of those agreeing with OpenModelica traces within our parity bands. On the subset of models both tools simulate and agree on, Rumoca reaches a runnable model several times faster, since it emits native code through a Cranelift JIT rather than generating C and invoking a compiler, while its interpreted integration loop is slower than OpenModelica’s native code on larger systems. That subset excludes the harder models by construction and skews small, so

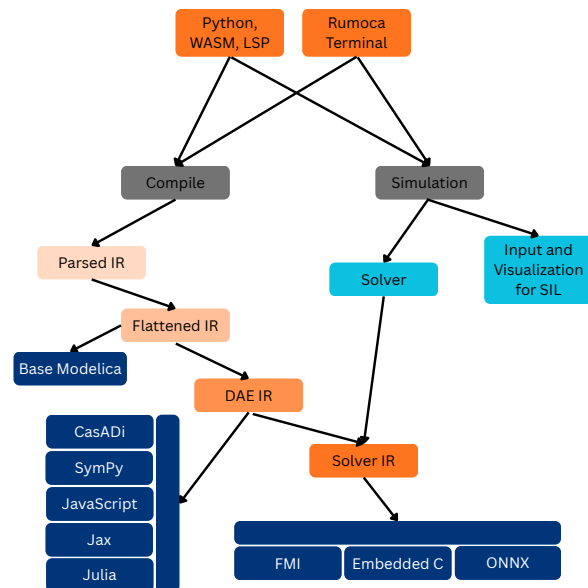


Figure 1. Rumoca’s architecture. Compile and Simulation are peer activities, each callable from the CLI, language server, Python, or WebAssembly frontends; the IRs feed the code generators and the native Rust runtime.

we offer the timings as evidence that the pipeline is fast enough to be practical, not as a general comparison against a production tool. The whole pipeline also builds to WebAssembly, giving a playground that runs client-side with no backend, and self-contained HTML simulators compiled from a single model.

Two end-to-end case studies close the loop. A 21-state quadrotor runs as a realtime software-in-the-loop plant for the CogniPilot Cerebri autopilot at 250 Hz, flown live through a gamepad against a browser-rendered scene. A twenty-line planar VTOL model is then emitted into four ecosystems from one source, where SymPy certifies that the CasADi and SymPy Jacobians agree to machine precision—a guarantee a hand-ported model cannot give.

Differentiable Parameter Optimization for DAEs with State-Dependent Events

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Abstract

Equation-oriented modeling tools routinely generate differential-algebraic equations (DAEs) with state-dependent events arising from switching logic, impacts, mode changes, and state reinitializations. Although mature DAE solvers robustly simulate such hybrid models, gradient-based parameter estimation remains difficult because algebraic variables are implicit, event times depend on model parameters, and resets are discontinuous. We formulate parameter learning for event-driven DAEs as constrained least squares and compare two complementary gradient strategies: automatic differentiation (AD) through the hybrid simulation and an explicit discrete-adjoint formulation based on an event-split trajectory.

We consider semi-explicit DAEs $\dot{x} = f(t, x, z, p)$, $0 = g(t, x, z, p)$ with state events defined by guards $\phi_e(t, x, z, p) = 0$ and reset maps relating pre- and post-event states. Both methods account for the DAE dynamics, event guards, resets, and post-event algebraic consistency. Their gradients are local to the event sequence selected by the forward simulation and assume transversal crossings with unchanged event identity and ordering under sufficiently small parameter perturbations.

The AD-through-simulation approach treats the hybrid simulator as a differentiable program. The algebraic variable is written implicitly as $z = \zeta(t, x, p)$ with $g(t, x, \zeta(t, x, p), p) = 0$, and its derivative is obtained from the implicit function theorem rather than by differentiating through nonlinear-solver iterations. State events are handled by segmented integration: a guard crossing terminates a segment, the reset is applied, algebraic consistency is restored, and integration resumes from the post-event state. Gradients therefore propagate through the algebraic solve, event-time localization, reset maps, and output reconstruction.

The discrete-adjoint approach targets external DAE solvers whose internal operations are unavailable to AD. The forward solver first generates an event-split trajectory containing smooth segments, event times, and pre- and post-event states. We then construct trapezoidal DAE residuals together with event residuals enforcing guards, resets, continuity, and post-event consistency. These residuals are equality constraints, not penalty terms. Their Lagrange multipliers define a backward discrete-adjoint

sweep that eliminates explicit trajectory and event-time sensitivities. Because event times define adjacent segment boundaries, their stationarity conditions are coupled explicitly to the neighboring smooth-segment residuals. This approach can be paired with mature external solvers and gives direct control over the adjoint discretization, although its accuracy depends on the residual mesh.

We evaluate both methods on a Cauer electrical ladder with five differential states, three algebraic variables, seven optimized parameters, and ten state-dependent reinitialization events over 20 s. From the same approximately $\pm 10\%$ biased initialization, both methods use 500 Adam iterations and the same observation grid. Both optimizers use the same sigmoid blending only for output reconstruction near event boundaries; the DAE dynamics, guard localization, and reset maps remain unsmoothed. Using a common IDA forward solve with hard, right-continuous event selection for evaluation, AD-through-simulation attains a loss of 1.32×10^{-6} compared with 4.01×10^{-4} for the discrete adjoint, at approximately 0.030 and 0.042 s per iteration, respectively. Several parameters are recovered to below one percent relative error by AD, while remaining errors indicate weakly identifiable parameter directions.

A planar bouncing-ball benchmark with $N \in \{3, 7, 15\}$ provides an additional ODE-with-events scaling study. On the tested single-CPU-thread implementations, JAX AD-through-simulation uses approximately $5\times$, $16\times$, and $37\times$ less total wall-clock time than direct PyTorch back-propagation as N increases. At $N = 15$, PyTorch reaches a lower validation loss (3.88×10^{-8} versus 1.35×10^{-5} for JAX AD), but at substantially greater computational cost. The discrete-adjoint implementation shows increasing bias as the fixed trapezoidal residual mesh becomes less adequate. This benchmark does not exercise the DAE algebraic solve.

The results highlight complementary deployment settings: AD-through-simulation is convenient when the simulator participates directly in differentiable programming, whereas the explicit discrete adjoint can use an external DAE solver and provides greater control over residuals and linear algebra. Future work includes scaling with DAE index, algebraic-loop size, tearing, and native Modelica or IDAS adjoints. Modelica/FMI integration and adaptive event-segment adjoint meshes are additional directions.

ModelScript: Rethinking the Digital Thread with a Web-Native, Polyglot Compiler for Modelica, SysML v2 & Beyond

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Designing complex cyber-physical systems demands continuous reconciliation and verification across heterogeneous, often siloed engineering artifacts, spanning system requirements, architectures, 3D CAD geometries, and multi-domain physical dynamics. While the Digital Thread promises to dissolve these silos by linking disparate tools and repositories, contemporary implementations cater almost exclusively to a high-latency, coarse-grained enterprise "Outer Loop." In this paradigm, cross-domain coordination relies on brittle, batch M2M conversions, passive database traceability links, and black-box co-simulations that are fundamentally ill-suited to an engineer's fast, interactive "Inner Loop."

To overcome these limitations, ModelScript reconceptualizes the Digital Thread's "Inner Loop" through an incremental, web-native compiler that runs client-side at the edge rather than in the cloud. The compiler maintains a code-centric Computable Polyglot Semantic Graph (CPSG) spanning heterogeneous engineering artifacts, including Modelica, SysML v2, and STEP AP242 CAD files. Cross-language projections are governed by hybrid Triple Graph Grammars (TGGs) evaluated over a demand-driven Salsa query engine. Meanwhile, flattening and linting are implemented as memoized Salsa queries, all expressed in a custom JavaScript-based DSL. By tracking fine-grained dependencies, ModelScript's Salsa query engine reactively invalidates only the query results affected by changed AST subtrees. It leverages Merkle hashes for value backdating, ultimately enabling responsive, low-latency compiler diagnostic feedback.

To power the Salsa query engine, ModelScript im-

plements an incremental GLR parser utilizing a zero-allocation, Data-Oriented Design (DoD) inside WebAssembly linear memory. This architecture packs AST nodes into compact 32-byte records that store 64-bit DJB2 Merkle hashes for $O(1)$ subtree dirty checking. For continuous, in-browser verification, an arena-based DAE builder transforms the flattener's implicit, high-index DAEs into explicit ODEs suitable for numerical simulation. The lowering pipeline then executes algebraic preprocessing, encompassing alias elimination, constant folding, and array scalarization, followed by index reduction, causalization, and algebraic loop tearing. The resulting ODE can be simulated directly in the browser using a WebAssembly-compiled SUNDIALS CVODE with SuiteSparse KLU, or exported as an FMU.

We demonstrate cross-language reconciliation and verification on a parameterized component battery pack that spans SysML v2 architectural constraints and Modelica physical dynamics. Although the development of ModelScript is ongoing, the compiler successfully flattens 670 out of 2,749 test models (24.4%) from the OpenModelica flattening test suite. It achieves fair coverage on foundational language constructs, including *extends* (76.8%), *equations* (66.7%), *types* (63.0%), and *modifications* (51.1%). Furthermore, empirical benchmarks across eight different edit conditions demonstrate that sub-300 ms diagnostic feedback is successfully maintained even when scaling up to $N = 5,000$ equations. By executing directly in the browser, ModelScript advances the Digital Thread from offline traceability to continuous verification at the edge.

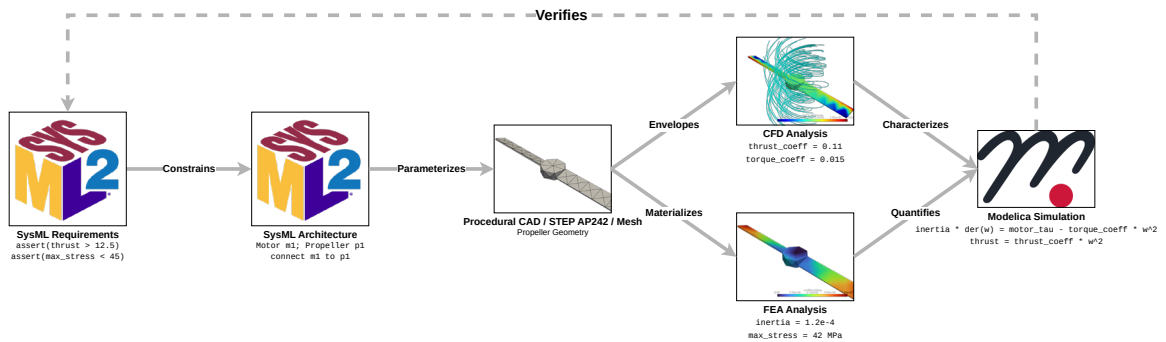


Figure 1. The Code-Centric Digital Thread: reconciling SysML v2 architecture, procedural CAD geometry, and Modelica physical dynamics with closed-loop requirements verification.

MMChvdcLibrary: Modeling Arm Average Value Modular Multilevel Converters in Modelica for Power System Simulation

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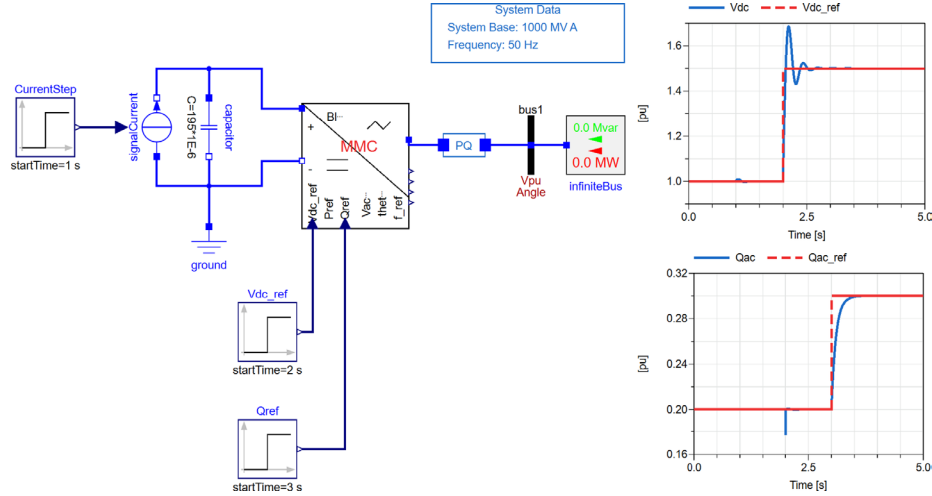


Figure 1. DCvoltageModeTest: the DCvoltageModeMMCStation between a DC-side current source and an AC grid equivalent (left); DC voltage tracking of the $V_{dc,ref}$ step (top right) and reactive power tracking of the $Q_{ac,ref}$ step (bottom right).

HVDC transmission based on Modular Multilevel Converters (MMCs) is the reference topology for modern HVDC links and future multi-terminal DC grids. Yet, electromagnetic transient models, accurate at switching level, can be prohibitively expensive to simulate and more importantly, to interpret, for grids with many AC buses, machines, and HVDC terminals, which motivates phasor-domain arm average value representations that preserve the essential average dynamics, including the converter's internal stored energy. A structured, multi-mode, openly available MMC-HVDC Modelica library has so far been absent.

What is presented. This paper introduces `MMChvdcLibrary`, a Modelica library developed in Dymola[®] for phasor-domain arm average value modeling of MMC-based HVDC systems (see Figure 1). The converter model preserves the internal energy dynamics of the arm capacitors through an equivalent capacitance $C_{eq} = 6C_{tot}$, and the control structure follows the energy-based framework, supporting constant and DC-voltage-linked energy references. The library is organized in packages: `ComponentLibrary`, which holds the converter model, inner and outer control loops, and other essential components; and `MMC_HVDCstation`, which assembles them into station models for P_{ac} - Q_{ac} , V_{dc} - Q_{ac} and V_{ac} phasor control, plus a user-configurable `MasterHVDCstation` that integrates all three behind a mode-selection input. It depends only on the Modelica Standard Library v4.1.0 and OpenIPSL v3.1.0, so stations drop into standard phasor-domain AC network models.

Results. Three example models demonstrate the library features, one per operating mode. To illustrate, in the `DCvoltageModeTest`, a DC-side current disturbance produces only a ≈ 0.02 p.u. excursion in V_{dc} , first absorbed by the arm capacitors and then removed by the controller; the subsequent $V_{dc,ref}$ step tracks with an underdamped response that settles about 0.9 s without steady-state error, with Q_{ac} regulated throughout. The other control modes are explained in detail in the paper.

Take-aways. The library offers a reusable, component-based framework for computationally efficient MMC-HVDC studies in Modelica, where a station is reconfigured for a different role rather than rebuilt. It is distributed under the BSD 3-Clause License at github.com/ALSETLab/MMChvdcLibrary. Ongoing work adds point-to-point and multi-terminal HVDC examples, black-start oriented test systems, and comparison against detailed EMT models.

Keywords: arm average-value model, modular multilevel converter (MMC), HVDC, power systems, Modelica, Dymola, OpenIPSL

Real-time Simulation and CHiL Testing of Power System Stabilizers on Microcontrollers with Modelica and eFMI

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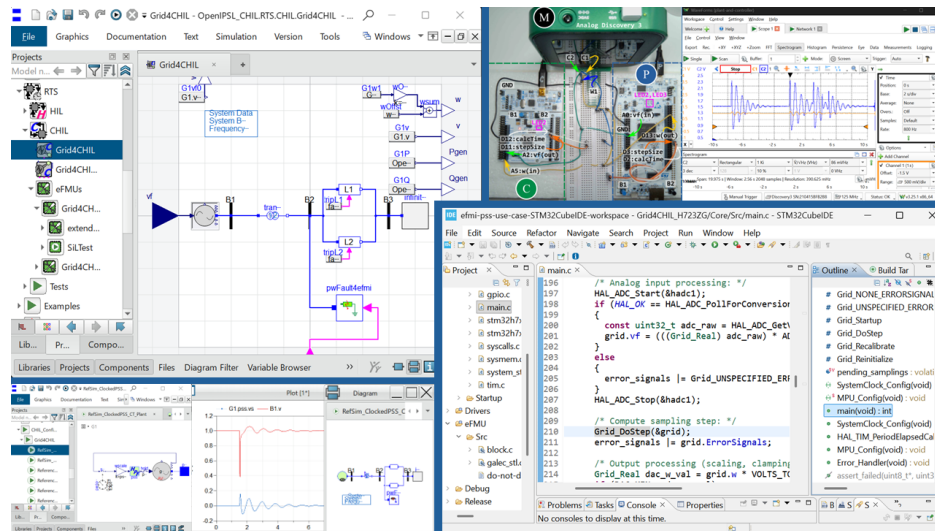


Figure 1. Top left & bottom right: Modelica model & derived eFMI STM32 firmware in Dymola & STM32CubeIDE; Bottom left & top right: MiL simulation of the CHiL setup & actual measurements in Digilent WaveForms.

Challenge. Modern power grids face significant stability challenges, as shown by recent incidents such as the 2025 oscillations of the Iberian grid, highlighting the need for robust model-based systems engineering (MBSE) to enable rapid control redesign and deployment. Yet, the move to controller hardware-in-the-loop (CHiL) testing is hindered by manual re-implementation, costly and complex tooling stacks, and the difficulty of obtaining real-time suited plant models—all of which disrupt traceability.

Presented solution. This paper pioneers a workflow using the Modelica[®] language and the eFMI[®] (Functional Mock-up Interface for embedded systems) Standard to bridge this gap in current practice within the power grid domain (see Figure 1). We introduce the OpenIPSL_CHIL Modelica library, extending the Open-Instance Power System Library (OpenIPSL) for embedded real-time applications: A clocked power system stabilizer (PSS) controller *and* the generator-with-grid plant become real-time-suited models, keeping physics as the single source of truth. The eFMI tooling of Dymola[®] in turn synthesizes both into production-ready C17 code (fixed 0.2 ms/5 kHz sampling, inline explicit-Euler integration), system-integrated using STM32Cube tooling for IO and clock-tree/timer configuration and automatic hardware abstraction layer (HAL) code generation, and deployed on two low-cost STM32[®] development boards with ARM[®] Cortex[®]-M microcontrollers wired in closed loop: A NUCLEO-L476RG (80 MHz) for the controller and a NUCLEO-H723ZG (550 MHz) for the plant.

Validation results. The workflow is validated by a full suite of model-, software- and controller hardware-in-the-loop (MiL, SiL, CHiL) tests. In CHiL time-domain tests the engaged PSS damps the generator speed and recovers from ± 0.2 pu deviations, whereas the uncompensated plant becomes unstable beyond ± 0.1 pu; measured open- and closed-loop frequency responses (0.15 Hz to 100 Hz) match the Dymola-linearized models. Non-functional quality is verified too: MISRA C:2023 and SEI CERT C Coding Standard compliance, strict eFMI GALEC error signaling excluding numerical issues, and measured execution times confirming real-time capability with ample headroom.

Take-aways. The workflow is a traceable and cost-effective alternative to proprietary real-time platforms, replacing setups typically costing $\gg \$50,000$ with $< \$50$ microcontrollers. All source models and generated artifacts are released under the 3-Clause BSD License at <https://github.com/ALSETLab/efmi-pss-use-case>, providing an open-source MBSE workbench for equation-based design, MiL-CHiL validation, and microcontroller deployment of power system stabilizers (PSS).

Keywords: power systems, power system stabilizer, real-time simulation, CHiL, embedded control, Modelica, eFMI, Dymola, STM32, ARM Cortex-M

Deploying OpenIPSL Models into DIgSILENT PowerFactory via FMI Model Exchange

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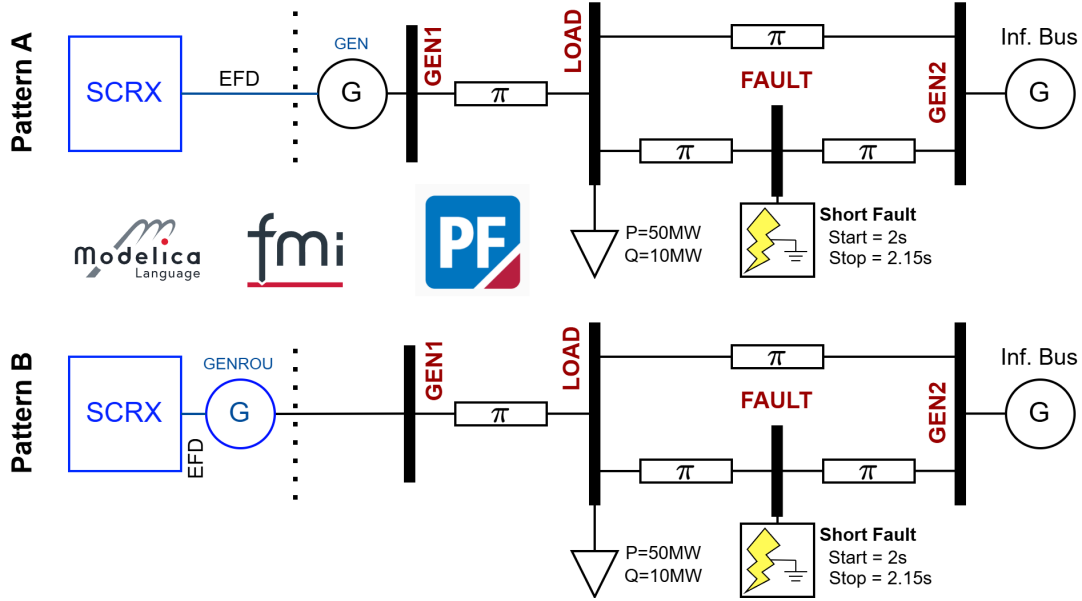


Figure 1. Two FMI Model-Exchange coupling patterns on the common benchmark: Pattern A, the OpenIPSL SCRX exciter as a controller-only FMU coupled to PowerFactory’s native synchronous machine; Pattern B, the OpenIPSL GENROU+SCRX as a full-machine & exciter FMU coupled to the grid through a voltage/power boundary.

Integrating open-source Modelica power-system models with commercial grid-simulation environments remains a practical barrier in utility workflows. The Open-Instance Power System Library (OpenIPSL) provides Modelica models for synchronous machines, excitation, and control system models, validated against PSS[®]E. Yet, utility studies are run in tools such as DIgSILENT PowerFactory, and to the authors’ knowledge, no published study documents an end-to-end deployment of Modelica, and in particular OpenIPSL models, into it.

What is presented. This paper documents a workflow using the Functional Mock-up Interface (FMI) in *Model Exchange* mode, in which PowerFactory’s RMS integrator advances the FMU’s states directly. OpenIPSL components are exported from OpenModelica as FMI 2.0 FMUs, registered as Modelica Model Types, and wired into PowerFactory’s Composite Model Frame. Two coupling approaches are defined and compared (see Figure 1): a *controller-only* pattern (Pattern A) that exchanges scalar AVR signals between the FMU and PowerFactory’s native synchronous machine, and a *full-machine* pattern (Pattern B) that couples the FMU to the grid through a voltage/power boundary.

Results. Both patterns are applied to a two-machine benchmark with the OpenIPSL GENROU and SCRX models and compared against PowerFactory’s built-in PSS[®]E SynMach SCRX under a 150 ms three-phase fault. All three configurations start from the same power flow (steady state), verifying the initialization conditions before the disturbance. Controller-only transients closely track the reference (RMS errors of 0.039 pu on EFD and 0.003 pu on V_t) at a runtime indistinguishable from the baseline. Pattern B, the full-machine coupling, matches in steady state, but shows a higher field-voltage peak (≈ 11 vs. ≈ 6.5 p.u.) and costs about $2.5\times$ the baseline simulation time. Pattern A exonerates the exciter model and FMI transport, leaving machine-level conventions and, most likely, the representation chosen at the machine-grid boundary, where a filtered power injection replaces a current source behind $R_a + jX''$.

Take-aways. Practitioners get a documented route for exercising validated Modelica models inside a commercial utility tool, with explicit trade-offs between the two coupling granularities. Workflow, FMUs and project files are open source at github.com/ALSETLab/Modelica-FMI-PowerFactory-ModelImport-Tutorial.

Keywords: Modelica, OpenIPSL, FMI, Model Exchange, PowerFactory, power system dynamics, model validation, synchronous machine, excitation system

Discontinuity-Aware Solvers for Vapor-Compression Cycles

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Physics-based vapor-compression-cycle models support heat-pump and air-conditioning design and control, but their simulation remains challenging. Finite-volume heat exchangers, refrigerant properties, and component constraints produce large, stiff differential-algebraic equation (DAE) systems. Refrigerant density is continuous yet has discontinuous derivatives at the bubble and dew boundaries, as illustrated for R32 in Figure 1. Adaptive integrators that assume smooth solutions may step across these boundaries, estimate large local errors, and enter clusters of rejected or very small accepted steps, substantially increasing simulation time.

We present a discontinuity-aware time-step controller for ODE and DAE solvers. Domain-specific guard functions identify discontinuity surfaces, and, unlike conventional event handling, the controller evaluates them after rejected as well as accepted trial steps. After a sign change, it locates the earliest crossing on the solver's dense interpolant, directs the next step toward that crossing, resets the controller history, and restricts the first subsequent step. The interpolant reuses rejected-step information, avoiding new full residual or Jacobian evaluations and matrix factorizations. Figure 2 shows how this logic eliminates a cluster of repeated rejections near a discontinuity.

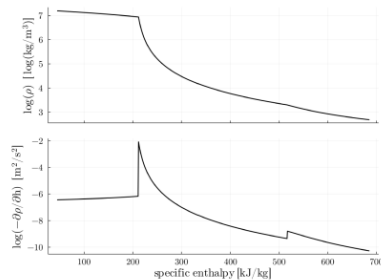


Figure 1. R32 density and density derivative at 1 MPa. The jumps mark the bubble and dew boundaries.

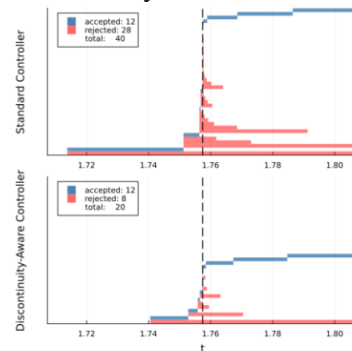


Figure 2. Trial steps near a discontinuity. Discontinuity awareness reduces rejected steps from 28 to 8.

The approach is natural for equation-oriented modeling. In a Dyad vapor-compression model, each refrigerant control volume declaratively marks its bubble and dew boundaries as functions of pressure and specific enthalpy. The compiler preserves this symbolic information while transforming the component model into a semi-explicit DAE and passes the guards to DifferentialEquations.jl. Model authors can therefore expose physically meaningful nonsmooth structure without reformulating or reassembling the equations, and the concept is applicable to Modelica-oriented workflows.

We evaluate the controller on a scalar ODE and a vapor-compression cycle using Tsit5, Rodas5P, and RadauIIA5. Figure 2 shows substantially fewer rejected steps in the scalar problem. For the full cycle, Rodas5P at loose tolerances gives about twofold average and up to eightfold tail speedups, including one reduction from 20 to 2.5 minutes.

Toward Equation-Based Building Energy System Modeling in Julia: A Modular Prototype and Verification Study

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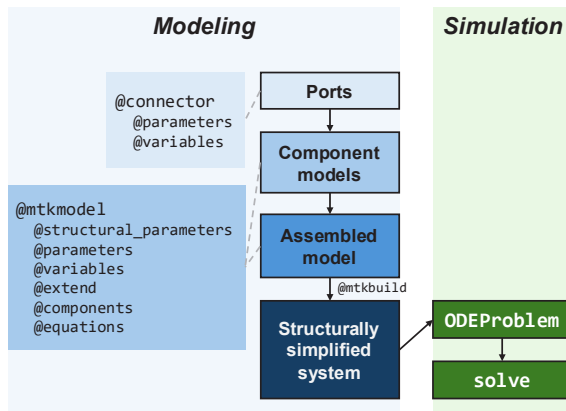
Abstract

A unified environment for equation-based modeling, optimization, and scientific machine learning makes Julia SciML a promising platform for building energy control applications. However, realizing the full potential of this emerging ecosystem may require domain-specific modeling prototypes that use modeling as the entry point to an integrated workflow. This paper presents an open-source, lightweight Julia equation-based modeling prototype comprising a set of modular weather, building thermal, and Heating, Ventilation, and Air-Conditioning (HVAC) components. The prototype is verified against reference models from the Modelica Buildings Library through single-zone cases of increasing system complexity, and the results demonstrate numerical agreement across the cases. The goal is to support future building-domain control-oriented and optimization-based workflows in the Julia SciML ecosystem.

Keywords: Equation-based modeling, HVAC, Building energy simulation, ModelingToolkit

Prototype and workflow

The prototype BuildEML (v0.1.0) is available at github.com/BE-HVACR/BuildEML.jl. In the presented workflow, physical models are built via *ModelingToolkit.jl* v10 as symbolic systems and organized into reusable hierarchical components. After the full system is assembled, it is transformed through automatic symbolic compilation and structural simplification, and then converted into an ODEProblem for numerical solution using *DifferentialEquations.jl* v7. The overall workflow is shown below.

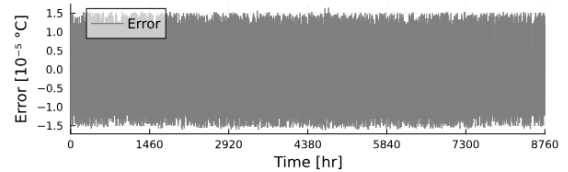
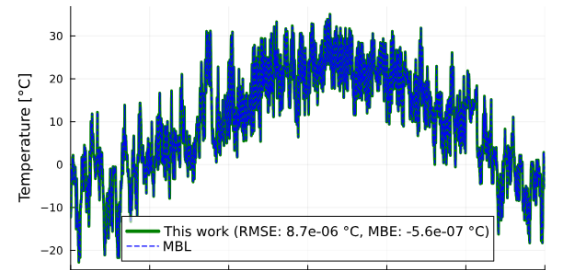


Verification results

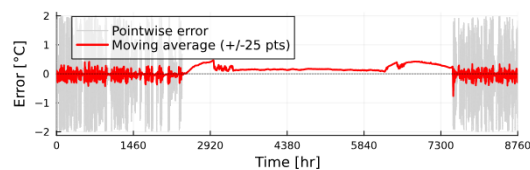
The verification compares four representative SimpleHouse variants in Modelica Buildings Library: SimpleHouse0 for weather-data processing, SimpleHouse3 for coupled envelope and zone-air dynamics, SimpleHouse5 for hydronic heating under hysteresis control, and the standalone SimpleHouse for integrated heating, cooling, and ventilation operation.

The SimpleHouse0 errors remain below 10^{-5} °C for outdoor temperature and 10^{-5} W/m² for global horizontal irradiance. The annual zone-temperature RMSE/MBE values are 0.01/-0.01 °C for SimpleHouse3, 0.54/-0.01 °C for SimpleHouse5, and 0.52/0.10 °C for the standalone case. Results of SimpleHouse0 and standalone SimpleHouse are shown below. The reasons for the observed differences, as well as the limitations and directions for future work, are also discussed in the paper.

Outdoor dry-bulb temperature (SimpleHouse0)



SimpleHouse, complete



Implementation of OpenIPSL in Dyad

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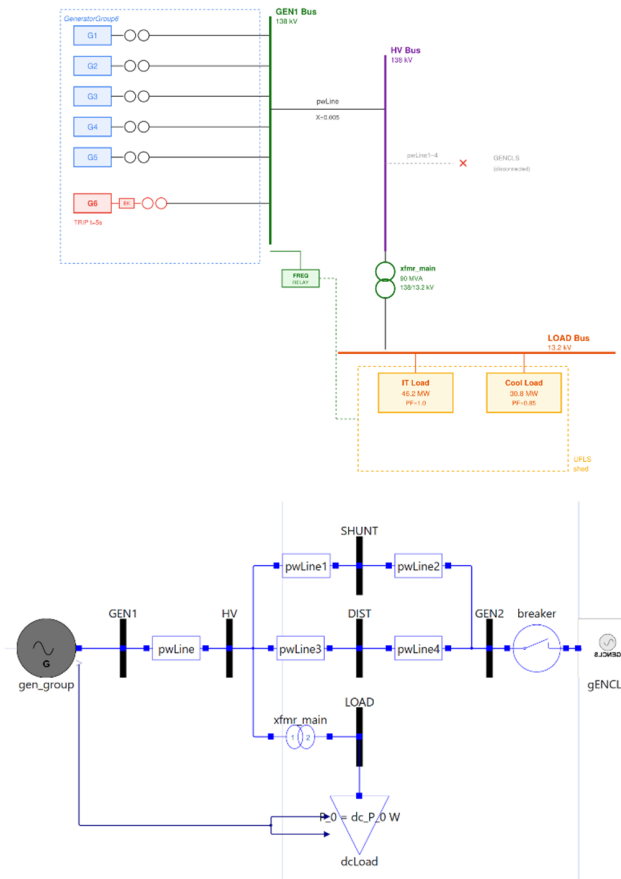
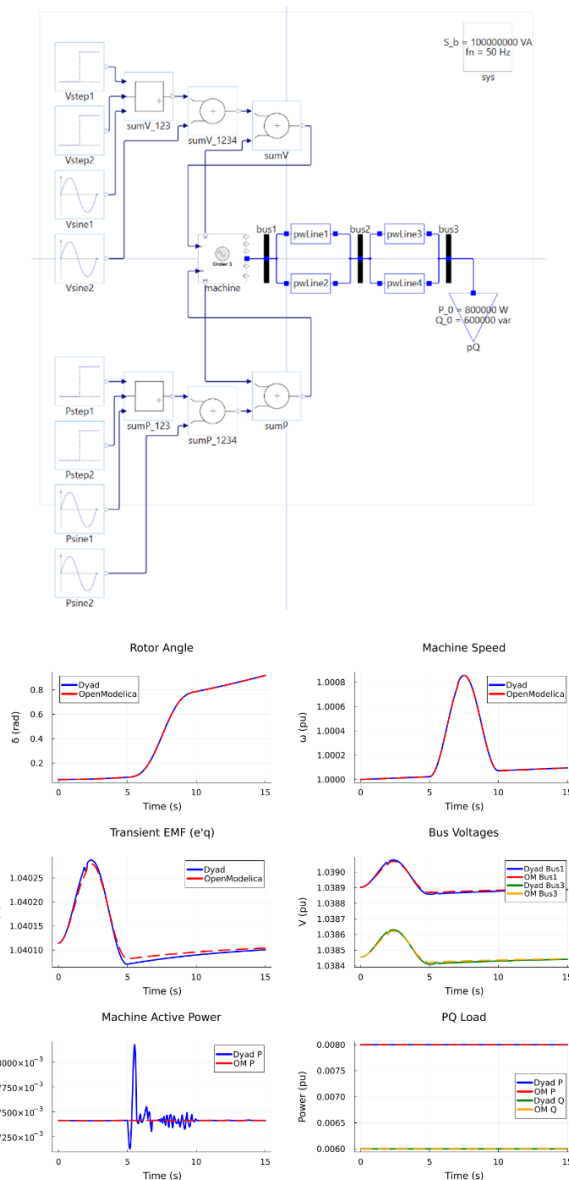
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Abstract

This paper presents a translation of the Open Instance Power System Library (OpenIPSL) into Dyad, an acausal modeling language that compiles to ModelingToolkit.jl in Julia. Details are provided regarding the translation process leveraging agentic AI capabilities and validation approach. The benefits of having both a system simulation specific AI agent and the library running in multiple tools are covered. The resulting library OpenIPSLComponents is also extended with additional components to demonstrate the potential for applications in data center power systems. Initial simulations for a gas turbine trip scenario are presented.

Keywords: power systems, phasors, data centers

Review OpenIPSL ElectricalLoads.PSAT.PQ in the resources folder. Propose an implementation in Dyad. Validate using OpenIPSL Tests.Loads.PSAT.LoadTestPQ.



Objective: Review the Dyad STS8 excitation system model, compare it against the original OpenIPSL Modelica source, and backport the bug fixes to Modelica.

Analysis performed:

Compared `dyad/Electrical/Controls/PSS/E5/r1b_model.dyd` (corrected Dyad implementation) against `resources/PSS/STS8.mo` (original OpenIPSL Modelica) and the PSS-E reference block diagram (`resources/PSS/STS8.png`). Identified three bugs in the original Modelica:

- `instantiateExcAgg` (and `lfi` block) used `T_dC1/T_dR1` instead of `T_dC2/T_dR2` — copy-paste error.
- `instantiateExcAgg` (and `CEL` block) used `T_dC1/T_dR1` instead of `T_dC2/T_dR2` — same copy-paste error.
- UEL/QEL lead-lag outputs were dead-ended — the main path (`instantiateExcAgg.y`) connected directly to the regulator gain (`Kr·ω`), bypassing any select logic. The UEL and QEL lead-lag chains computed outputs that went nowhere.

Files created:

File	Description
<code>resources/PSS/STS8_corrected.mo</code>	Corrected STS8 model with all three fixes applied. Adds a <code>SelectLogic3</code> instance to route the three lead-lag path outputs to the regulator.
<code>resources/PSS/SelectLogic3.mo</code>	New OpenIPSL <code>NonElectricalLogicalSelectLogic3</code> block implementing the three-way selection logic from the IEEE/PSS-E reference. QEL active → V1, UEL active → V2, otherwise → V3.

No Dyad files or generated code were modified.

A Single-Element Multi-Point Compliance for Rigid-Body Connections

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Abstract

Many engineering assemblies connect two rigid bodies at several mounting points through compliant elements, such as automotive subframes joined to the body via bushings. Modeling each mount as an independent 6-DOF bushing is intuitive but couples many stiff parallel force paths into the rigid-body equations, introducing kinematic loops and severe numerical stiffness that degrade robustness and speed. This abstract summarizes a single-element multi-point compliance (SEMPC) that aggregates an arbitrary set of local translational and rotational stiffnesses into one 6×6 spatial stiffness at a body reference frame. The formulation is first-order accurate in relative orientation, adds no algebraic complexity over an ordinary point bushing, and provides a linear map that redistributes the net wrench to individual mounts for sensing. A second-order filter in wrench space preserves stability at high stiffness and real-time step sizes.

Keywords: Multi-point compliance, MultiBody Dynamics, Spatial stiffness

Problem and Approach

Multiple compliant attachments between two rigid bodies are ubiquitous: subframe-to-chassis interfaces, engine mounts, exhaust hangers, and control-arm bushings. The conventional approach instantiates one 6-DOF bushing per attachment. Treated as pure force elements, the high stiffness of automotive mounts drives dominant free-response frequencies into the kilohertz range, forcing impractically small explicit step sizes. Treated with the deflection selected as a state, several elements sharing the same body pair can contribute only one independent deflection; the rest are tied by nonlinear loop-closure constraints on $SO(3)$. Both formulations therefore penalize robustness and efficiency.

The proposed SEMPC pre-assembles a single 6×6 spatial stiffness $\mathcal{K}$ from the local stiffnesses $\mathbf{K}_i, \mathbf{K}_{\theta,i}$ and attachment geometry $\mathbf{r}_{i/a}$, and computes the net wrench from the relative pose $[\Delta \mathbf{r}; \Delta \theta]$ at a chosen reference frame:

$$\begin{bmatrix} \mathbf{F} \\ \mathbf{M} \end{bmatrix} = \mathcal{K} \begin{bmatrix} \Delta \mathbf{r} \\ \Delta \theta \end{bmatrix},$$

with the aggregate stiffness assembled from the mount

contributions as

$$\mathcal{K} = \sum_{i=1}^n \begin{bmatrix} \mathbf{K}_i & -\mathbf{K}_i [\mathbf{r}_{i/a}]_{\times} \\ [\mathbf{r}_{i/a}]_{\times} \mathbf{K}_i & \mathbf{K}_{\theta,i} - [\mathbf{r}_{i/a}]_{\times} \mathbf{K}_i [\mathbf{r}_{i/a}]_{\times} \end{bmatrix}.$$

The multi-point assembly is thus equivalent, to first order, to a single 6-DOF element with a precomputable stiffness, free of extra algebraic unknowns and kinematic loops. A companion load-splitter map $\mathcal{A}_i \mathcal{K}^{-1}$ recovers per-mount wrenches for diagnostics without adding states, and a second-order (inertance-damper-spring) filter in wrench space lowers the effective natural frequency to stabilize explicit fixed-step integration.

Modelica Implementation and Results

The SEMPC is realized as a Modelica MultiBody component with `frame_a/frame_b` connectors, combining a passive `MultiMountSensor` (algebraic compliance and per-mount load observer) with a `VariableTranslationRotation` actuator that integrates the second-order filter. Both $\mathcal{K}^{-1}$ and the load-splitter matrices are `final` parameter constants, so no factorization is required at runtime and only twelve scalar ODE states remain.

A benchmark against a conventional model of independent 6-DOF spring-dampers, using explicit fixed-step Euler in Dymola, shows the reference model is stable only at $h_{\text{ref}} = 1 \times 10^{-7}$ s (about $40\times$ slower than real time), while the SEMPC is stable at $h_{\text{SEMPC}} = 1 \times 10^{-3}$ s—four orders of magnitude larger—and runs about $25\times$ faster than real time while closely tracking the reference displacement and per-mount loads. Because the runtime dynamics reduce to a single fixed-state 6-DOF element, SEMPC wall-clock time is nearly independent of the number of mounts, whereas the reference cost grows with n . The full paper details the derivation, the Modelica component, and the case studies.

A 2-DOF MultiBody Track Joint for Race Car Optimization

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Abstract

TrackJoint is a reusable Modelica MultiBody component that represents a three-dimensional racetrack as a two-degree-of-freedom kinematic joint. Longitudinal arc length and lateral offset determine the position, orientation, and angular velocity of a road-attached frame. Additional joints retain the chassis heave, roll, pitch, and yaw motions. Periodic piecewise-quintic splines supply the road geometry, avoiding separate tables of road angles. The same vehicle-and-track model is used without modification in Dymola time-domain simulation and, through FMI 3.0 export, CasADi/IPOPT minimum lap-time optimization. Road-relative chassis coordinates also simplify initialization of a full-lap problem. Validation against LiDAR-derived terrain and a converged Singapore lap-time solution demonstrate both uses.

Unlike a one-dimensional rail joint, **TrackJoint** permits both motion along a reference line and lateral motion across the road (Figure 1). Six periodic quintic splines describe the reference line and lateral elevation polynomial

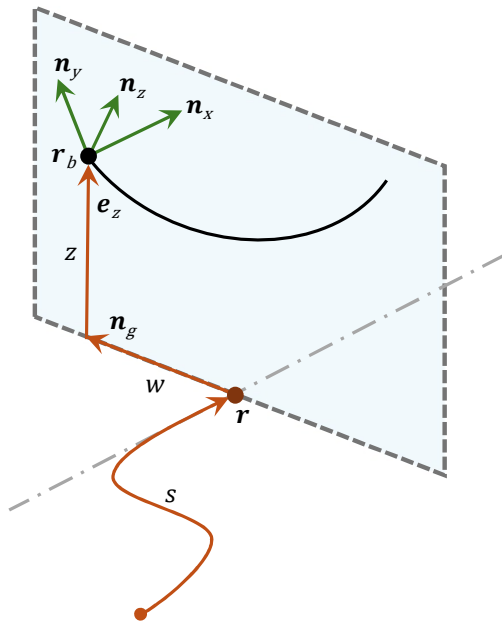


Figure 1. **TrackJoint** constructs the surface point r_b and its orientation from longitudinal position s , lateral offset w , and road elevation z . The attached frame follows the surface with two free coordinates.

of a Monge-patch surface. Their C^4 continuity supports the surface derivatives required by Modelica index reduction. The joint constructs the surface frame directly from those derivatives and closes the road constraint through d'Alembert's principle, without explicit Lagrange multipliers.

The standard `frame_b` connector supports a heave joint and three track-relative rotational joints. The chassis therefore retains all six degrees of freedom while its coordinates are aligned with the road. The component consumes fitted track data; the companion track-fitting paper addresses the choice of a reference line without normal crossings inside the drivable region.

Vehicle and joint export as one FMI 3.0 model-exchange FMU. The continuous state s and its derivative $\dot{s}$ supply the time-to-distance transformation used by FMIOPT to formulate a CasADi direct-collocation problem. Setting the track-relative chassis coordinates to zero and ramping s over the lap provides a road-aligned, kinematically consistent initial guess without a preliminary simulation. This satisfies the road geometry by construction; IPOPT must still resolve the vehicle dynamics and remaining optimization constraints.

In Dymola 2026x Refresh 1, a point mass driven along the Singapore reference line reproduces LiDAR-derived terrain height to within 6 cm over the lap (Figure 2). Grade and bank are also compared with the measured terrain. For the detailed multibody vehicle, IPOPT converges from the road-aligned guess in 264 iterations to a 90.65 s lap.

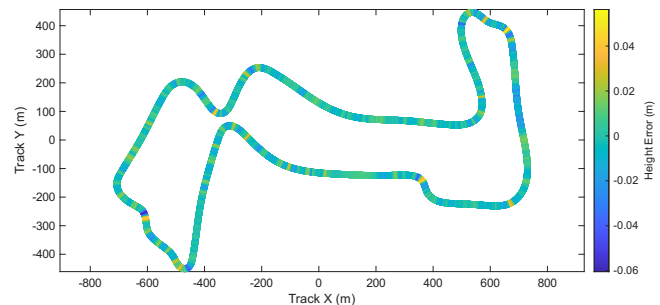


Figure 2. Height error against the LiDAR-derived Singapore terrain along the point-mass validation path. Absolute height error remains within 6 cm over the lap.

A Singularity Avoiding Track Model for Race Car Optimization

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Abstract

This work presents a pipeline that converts measured racetrack point clouds into closed, C^4 -continuous three-dimensional surfaces for high-fidelity Modelica vehicle simulation and minimum lap-time optimization. The surface is a Monge patch parameterized by arc length along a minimum-curvature reference line and signed lateral offset. A geometric centre-line can produce crossing lateral sections inside tight corners, compromising the road coordinates and the conditioning of the optimization problem. The pipeline computes a minimum-curvature line and checks that the reconstructed lateral sections do not intersect within the track. On the Singapore circuit, the resulting geometry enables convergence of an FMI-based CasADi/IPOPT minimum lap-time problem that fails with the centre-line reference.

An iterative quadratic program computes the minimum-curvature line subject to track bounds and periodicity, with jerk regularization. Recomputed widths and lateral sections are checked separately for non-intersection (Figure 1). The reference is not a prescribed driving trajectory.

Lateral polynomials fit surveyed elevations in the new frame. Reference coordinates, elevation coefficients, and

widths are exported as periodic quintic splines on shared knots. Their C^4 continuity supports surface derivatives and Modelica index reduction in the TrackJoint component.

The pipeline fits a circuit in roughly five minutes, reproducing surveyed elevations and boundaries within a few centimetres. Vehicle and joint export as one FMI 3.0 FMU for FMIOPT's CasADi/IPOPT optimization. On Singapore, replacing the centre-line with the minimum-curvature reference enables convergence without vehicle or solver changes. Figure 2 shows the optimized speed and solver histories.

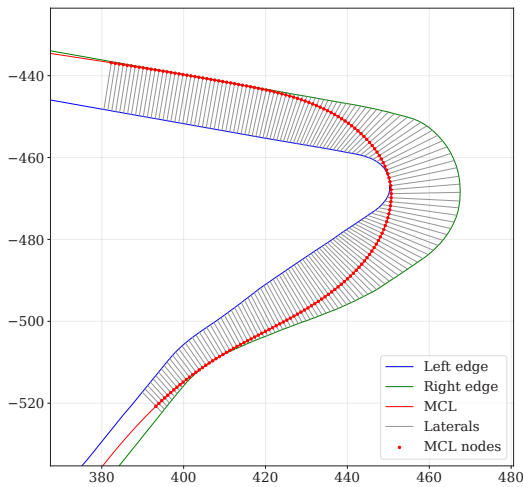


Figure 1. Minimum-curvature reference (red) and reconstructed lateral sections through a tight corner. Non-intersection is checked within the drivable region.

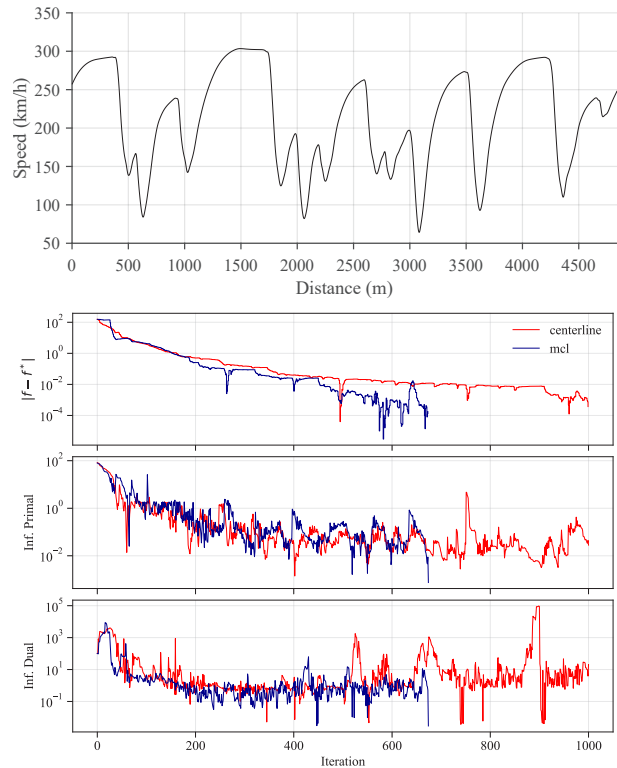


Figure 2. Singapore optimization results: optimized speed trace (top) and IPOPT iteration histories for centre-line and minimum-curvature references (bottom).

Five-Link Rear Suspension for F1: A Two-Stage Agentic AI Study with the Vehicle Dynamics Library

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Formula 1 teams must use a largely fixed chassis across circuits that place very different demands on mechanical grip, aerodynamic stability, wheel travel, and kerb compliance. This paper investigates whether a five-link rear suspension provides a broader kinematics-and-compliance (K&C) tuning range than a conventional double-wishbone, and whether that additional freedom can help one chassis accommodate circuits of opposing character. The study uses an agentic AI workflow connected to the Modelon Vehicle Dynamics Library (VDL) and Modelon Impact. Because VDL does not include a ready-made five-link rear linkage, the agent assembled one from validated suspension primitives, integrated it with the standard K&C test rig, compiled and debugged the model, and automated the parameter sweeps and post-processing. The engineering hypothesis, circuit targets, and final interpretation remained subject to human definition and approval.

The comparison is organized as a two-stage study. First, 500 F1-representative configurations are evaluated for each topology by varying chassis- and upright-side pickup locations and link compliance. A ± 50 mm wheel-travel sweep and lateral loading up to 4 kN are used to extract four descriptors: mean camber slope S_γ , camber curvature C_γ , mean toe slope S_τ , and toe curvature C_τ . The double-wishbone, formed by two A-arms and a tie rod, constrains several pickup points to common arm planes. The five-link replaces the A-arms with four independently placed binary links and retains the tie rod, giving greater freedom to shape the wheel-carrier motion. Across both stiff and compliant settings, the double-wishbone responses remain tightly clustered, while the five-link produces substantially broader camber and toe envelopes. This advantage persists as lateral load increases, indicating that compliance changes shift the double-wishbone response but do not open comparably large new regions of K&C behavior.

The second stage compares these reachable envelopes with illustrative targets for Monaco and Suzuka. Monaco represents a broad travel envelope, predictable response, and some compliance for frequent kerb interaction; Suzuka represents higher camber gain and stronger platform control under sustained aerodynamic load. At 4 kN, the compliant five-link setting ($k = 10^7$ N/m) overlaps the Suzuka target, while the stiff setting ($k = 10^8$ N/m) overlaps Monaco. The double-wishbone reaches the Monaco camber region but not the Suzuka camber region, and its compliant variant falls outside the Monaco toe target. The target boxes are based on qualitative circuit reasoning rather than measured vehicle optimization, so the result should be interpreted as a demonstration of the method and of the structural difference between the topologies, not as a quantitative performance prediction.

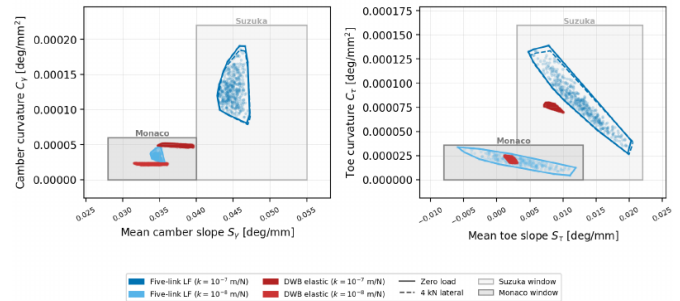


Figure 1. Indicative Monaco and Suzuka targets overlaid on the topology design windows at 4 kN lateral load. The two five-link compliance settings reach both target regions; the double-wishbone does not.

The results suggest a useful division between design-time and event-level decisions. Pickup hardpoints establish the season-long operating envelope because chassis and upright brackets are largely fixed, whereas calibrated link bushings or bearing variants provide a practical between-event compliance adjustment. More broadly, the work illustrates where an agentic workflow adds value: a hardened, composable physical library supplies trustworthy components, while structured tool access allows the agent to navigate unfamiliar models, create experiments, and preserve traceable results much faster than a manual workflow. The agent's contribution is operational rather than physical; engineers still define the question, review generated code, approve assumptions, and sign off on conclusions. With measured circuit targets and a validated baseline vehicle, the same approach could support quantitative season-wide suspension design decisions.

Keywords: Modelica, agentic AI, vehicle dynamics, five-link suspension, Formula 1, kinematics and compliance, Modelon Impact, Vehicle Dynamics Library

A Modelica-based Virtual Testbed for Hybrid-Cooled Data Centers

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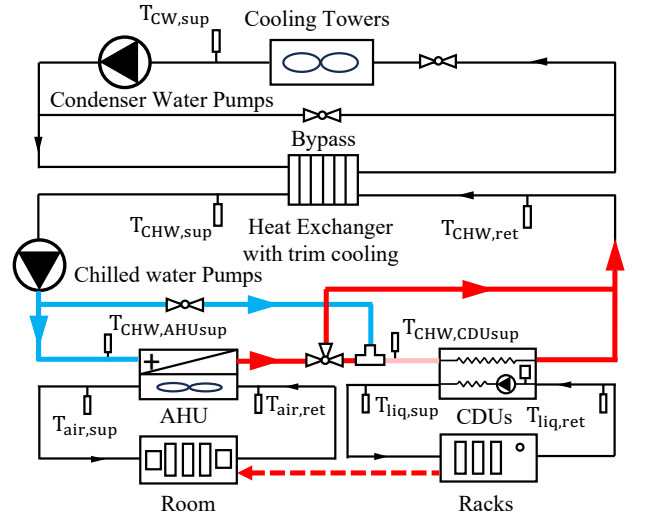
Abstract

Hybrid cooling systems that integrate air and liquid cooling are increasingly adopted in data centers to manage high-density and spatially nonuniform thermal loads. In these systems, the air- and liquid-cooling subsystems are dynamically coupled and exhibit different responses to variations in IT load, making coordinated control challenging. However, models capable of capturing the interactions between air- and liquid-cooling systems remain limited in the literature. To address this gap, this paper develops a hierarchical Modelica-based model spanning the system and rack levels. The model builds on standard components from the Modelica Buildings Library while introducing a hierarchical, physics-based representation of hybrid data-center cooling that explicitly captures plant-level coupling and dynamic rack-level heat partition between the liquid- and air-cooling paths. At the plant level, the air-handling unit and coolant distribution units share a chilled-water loop. At the rack level, a reduced-order thermal network dynamically partitions IT heat between the liquid- and air-cooling paths and couples the rack to the surrounding room air.

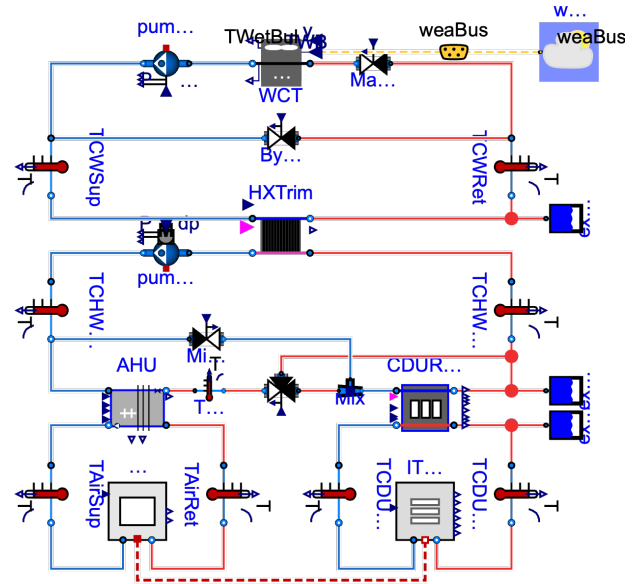
The model is validated using open-source operating data from the Energy System Integration Facility at the National Laboratory of the Rockies. Validation is performed for liquid-side heat gain, air-side heat gain, return air temperature, and return liquid temperature. The resulting NMBE values are -1.12% , 4.76% , 4.62% , and 0.95% , while the corresponding CVRMSE values are 2.68% , 11.31% , 5.29% , and 2.35% , demonstrating good agreement with measured system behavior.

To demonstrate its use for supervisory control, the validated Modelica model is exported as a co-simulation FMU and coupled with a Python-based grid-search algorithm. The controller coordinates air- and liquid-side supply-temperature setpoints while retaining local PI control. Compared with fixed-setpoint operation, the optimized strategy maintains thermal constraints and reduces total cooling energy consumption by 8.3% , including a 19.9% reduction in chiller energy. The results show that the proposed virtual testbed provides a transparent and reusable platform for studying coupled thermal dynamics and coordinated control in hybrid-cooled data centers.

Keywords: Modelica, Virtual Testbed, Data Center Cooling, Hybrid cooling, Optimization



(a) Hybrid cooling system schematic



(b) Modelica-based virtual testbed

Figure 1. System-level schematic and corresponding Modelica implementation of the hybrid-cooled data center.

Integrated Microclimate Model in Modelica for Data Center Chiller Farms

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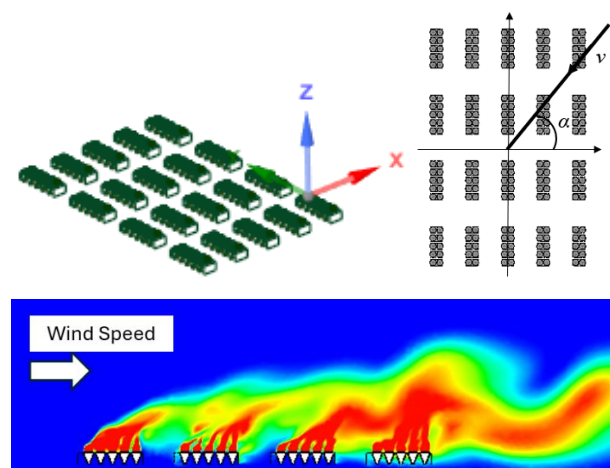
Trane Technologies, USA

Keywords: data center cooling, air cooled chiller, dry coolers, microclimate, Gaussian plume, reduced order model

Data centers are increasing in cooling density while also facing tighter energy and operational flexibility requirements. Air cooled chiller plants are attractive for many data center applications because they reduce or eliminate water use and can be deployed modularly, but closely spaced chillers can experience hot condenser exhaust recirculation. This recirculation raises local condenser entering air temperature, reduces coefficient of performance, and can create unequal capacity and efficiency penalties across a chiller array.

$$T_{CondIn} \uparrow \rightarrow COP_{chiller} \downarrow$$

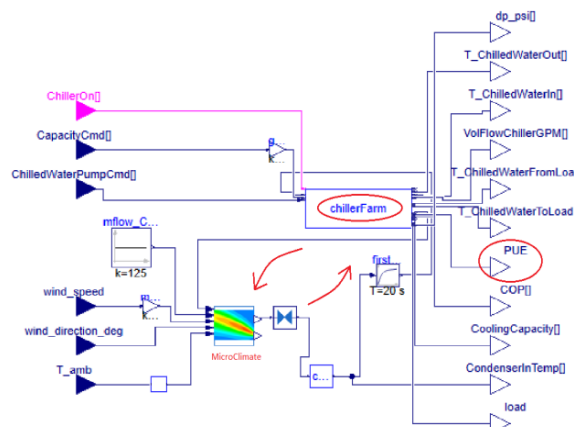
Computational fluid dynamics (CFD) remains the preferred tool for detailed airflow design studies, but repeated CFD simulations are computationally expensive for long horizon weather simulation, sequence evaluation, and control co-simulation. This paper presents a reduced order Gaussian plume-based microclimate model implemented in Modelica and coupled with a data center chiller farm model.



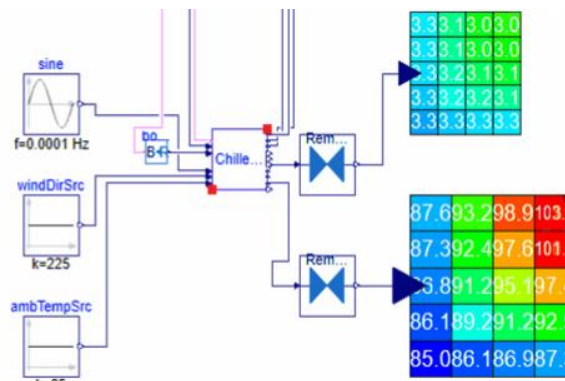
The chiller farm supplies heat rejection and condenser air flow for every operating unit. Together with ambient temperature, wind speed, and wind direction, the microclimate block estimates a local condenser inlet temperature for each chiller. A sensible energy balance estimates source exhaust temperature rise. Wind aligned downwind and crosswind distances describe transport and decay, while a buoyancy term represents the tendency of warm exhaust to rise away from nearby condenser inlets. Contributions from all operating chillers are superimposed at each target chiller.

The model is calibrated using 65 computational fluid dynamics cases for a 4x5 array of 300 ton air cooled chillers. Six parameters represent baseline recirculation, downwind and crosswind decay, calm condition decay, buoyancy influence, and the transition between calm and wind active behavior. Separate cases are retained for validation so that calibration improvement is not interpreted as predictive accuracy.

Gang Wang
Zhanwei He
Zhong Guo



An illustrative dynamic simulation demonstrates why the microclimate matters at plant level. With ambient air at 85.0 °F and wind speed reaching approximately 16 m/s, the hottest downstream condenser inlet reaches 102.8 °F. This is a local temperature rise of 17.8 °F above ambient. Across the chiller array, coefficient of performance ranges from approximately 3.4 at a favorable location to 3.0 at the most affected location, a difference of about 11.8%. A conventional plant model that applies one ambient temperature to every chiller cannot represent this spatial penalty or its influence on equipment sequencing.



This model does not replace detailed computational fluid dynamics for final chiller yard design. Current location dependent validation errors remain higher than desired, reflecting near field fan interaction and site geometry that a compact plume formulation cannot fully resolve. The model is intended for rapid screening, recirculation diagnostics, long horizon weather simulation, sequence studies, and development of recirculation aware supervisory controls.

Key contribution: spatially resolved condenser inlet conditions become available directly within fast Modelica plant and control simulations.

Agent-Assisted Construction of Modelica System Models from Validated Components: A Data Center Cooling Case Study

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Keywords: *Modelica, agentic AI, Model Context Protocol, data center cooling, validated component libraries, AI-assisted modeling*

How long does it take to go from a vendor reference design PDF to a simulation-ready Modelica model? For an experienced engineer, typically three to four working days. In this paper, an AI agent does it in approximately four hours — from a single user prompt, with no manual coding.

The agent operates through Modelon Impact using MCP tools for library access, Modelica generation, simulation, and KPI extraction. Before model assembly, all extracted and assumed engineering values are presented to the engineer for review — making every assumption visible and correctable before any Modelica is written. The study used Claude Sonnet 4.6 (Anthropic); the framework is LLM-provider-agnostic.

The task is to build a simulation-ready 7.4 MW dual-loop model conforming to Schneider Electric RD108, the Tier III AI reference design for NVIDIA GB200 deployments, starting from a validated 1 MW reference. RD108 is not a scaled-up version — it is a fundamentally different topology requiring two independent cooling loops: a high-temperature circuit at 37 °C for AI rack CDUs and a low-temperature circuit at 23 °C for networking racks, each with independent chillers, cooling towers, and control strategies.

The agent extracts 27 engineering parameters from the 10-page RD108 PDF, queries the DataCenter library for matching component records, and writes 405 lines of Modelica composing the dual-loop topology. It compiles and runs a 24-hour Berlin TMY dynamic simulation. All but one KPI pass on the first run. The LT loop supply temperature reads 42 °C against a 23 °C target — a 19 °C shortfall. The agent traces the thermal balance, identifies the LT chiller record as undersized by a factor of 1.7, re-queries the library, substitutes the correct 1,881 kW record, and re-runs. All KPIs pass (Table 1). The corrected model is 366 lines — cleaner than the first attempt.

Table 1. Final validated KPIs for the 7.4 MW model against RD108 targets, 24-hour Berlin TMY dynamic simulation.

KPI	Target	Result	Status
PUE	1.10	1.153	Close
P_{T} total	7,392 kW	7,392 kW	Pass
$T_{\text{FWS,HT}}$	37 °C	41.1 °C	Close
$T_{\text{FWS,LT}}$	23 °C	23.0 °C	Pass
$\dot{Q}_{\text{CDU}}$	5,512 kW	≈5,500 kW	Pass

What makes this reliable is not the agent alone. The Data Center Library provides pre-validated component records — chillers, CDUs, cooling towers, racks — parameterised from publicly available manufacturer specifications and verified against reference performance data. The agent selects and composes; the library guarantees the physics. When the first build fails, the agent corrects it by re-querying the same library. There is nothing to debug that the library does not already know how to answer.

This division of labour is the key insight: the agent handles systematic, time-consuming derivation and assembly work, while validated library components carry the physics that would otherwise require domain-expert modelling. The approach generalises wherever a validated library covers the target domain, the reference design is well-specified, and the KPIs are machine-checkable — all of which hold for a large class of building and facility applications.

The talk will walk through the complete workflow live: parameter extraction, library queries, model assembly, the diagnosis episode, and the corrected result. The case study shows that validated Modelica libraries can now be used not only for model construction, but also for automated diagnosis and correction — turning decades of accumulated library knowledge into an active engineering resource.

A Unified Dynamic System Testing Framework

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DSTF is open source under an MIT license on GitHub:

<https://github.com/ORNL-Modelica/DynamicSystemsTestingFramework>

Abstract

Regression testing of time-dependent behavior recurs in every simulation ecosystem, yet each has rebuilt a similar harness tied to a single backend, a single metric, and a single reference baseline: tube-comparison scripts guarding Modelica libraries, per-project FMPy drivers over FMUs, hand-coded trajectory checks in Julia, recorded experimental data reshaped into the same pass/fail pipeline. All share a structure their interfaces do not expose — a source of tests, a backend, a trajectory, a baseline, and a rule that decides "passed" — and none of it survives the crossing between ecosystems.

The **Dynamic Systems Testing Framework (DSTF)** is an open-source framework organized around six plug-in layers: Source, Discovery, Backend, Dataset, Metric, and MetricTree. Four are extended through a registry, so a new simulator, in-source test convention, or scoring rule arrives as a plug-in rather than a fork of the harness.

Two ideas carry the work. First, **acceptance criteria compose**. A test is scored by a tree of metric leaves — NRMSE, tolerance tube, checkpoints, value range, event timing, FFT peaks, scalar reductions — under roll-up operators ('AND', 'OR', 'k-of-N', 'weighted', advisory 'warn'), each leaf scopable to a time window, so one test can demand a faithful start-up transient, a steady-state envelope, and correct event timing at once. Second, **baselines carry an enforced role**: a *primary* anchor the test must score against, a *'soft check'* advisory reference such as a second simulator that is reported but can never fail the test, and a plot-only *companion* such as an experimental dataset.

Criteria are authored through a machine-readable contract rather than a particular editor: DSTF exports the JSON Schema its specifications validate against, and every edit arrives as an RFC 6902 JSON Patch validated as a whole tree before it is written. The HTML report is one client; a script or an AI assistant is another.

DSTF currently has been tested on five backends (Dymola, OpenModelica, FMPy, Julia/ModelingToolkit, Python), on four bundled demonstration libraries, and on the TRANSFORM Modelica library, whose suite of more than three hundred tests it now carries. The Python backend needs no simulator at all: one example returns the columns of a CSV recorded on a test bench, scored exactly as a live Dymola run is.

Keywords: regression testing, Modelica, FMI, Julia, validation

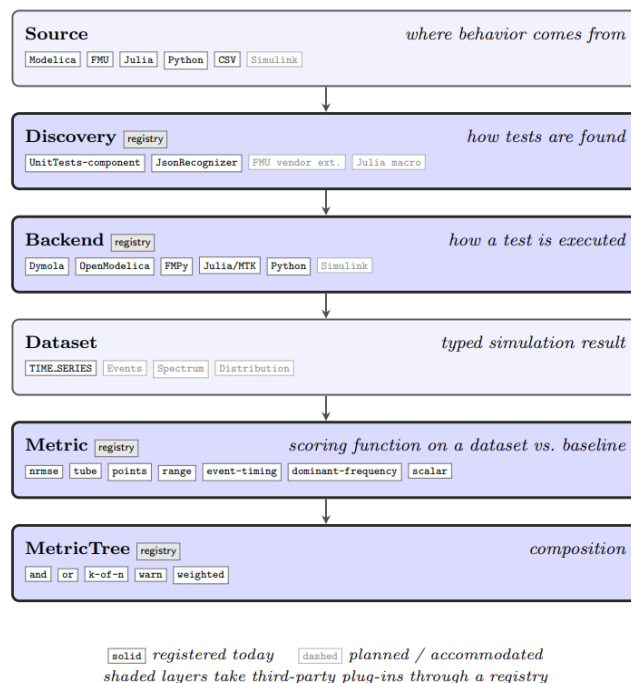


Figure 1. The six-layer plug-in architecture of DSTF. Every layer has a defined interface, and new recognizers, simulators, metrics, or combinators install into the registry of their layer without disturbing the others.

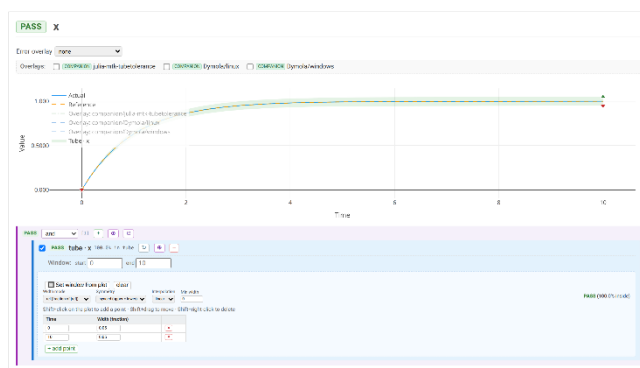


Figure 2. DSTF's report with a *tube* leaf's editor open. The control points defining the envelope are listed in the table and editable.

Data-Driven Refinement of a Native Modelica Model Predictive Controller for Multi-Variable Gas Turbine Hydrogen Blending Control - Abstract

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Abstract

As gas turbine operators increasingly pursue hydrogen and hydrogen-blend fuels to decarbonize power generation, the need for advanced multi-variable control strategies that can safely manage the complex interactions between fuel composition, combustion dynamics, emissions, and power output has become critical. Hydrogen's lower volumetric energy density, wider flammability range, and faster flame speed fundamentally alter the combustion tuning landscape, requiring simultaneous coordination of fuel flow, inlet guide vane position, and fuel staging. These tasks may exceed the capability of traditional single-loop PID architectures. Model Predictive Control (MPC) offers a principled framework for this coordination, but its implementation within equation-based system modeling environments such as Modelica can be challenging per the authors experience, particularly for high-fidelity, physics based gas turbine plant models where nonlinearity, stiff dynamics, and actuator constraints create significant practical challenges.

This paper extends the authors' prior work (Harper 2026), which demonstrated the feasibility of a fully native Modelica MPC architecture including Kalman filter state estimation, prediction horizon matrices, and a quadratic programming solver implemented entirely within the Modelica language for simultaneous control of hydrogen blend ratio, generator load, and turbine firing temperature. Where the prior paper established the architectural framework and initial proof of concept, the present work focuses on the systematic, data-driven refinement of both the internal plant model and the constrained optimization solver required to achieve robust closed-loop performance when coupled to a nonlinear, physics-based gas turbine simulation.

Key contributions include an updated open-loop system training data and identification procedure and the active set quadratic programming solver was reformulated to employ explicit Lagrange multiplier computation through a full Karush-Kuhn-Tucker (KKT) system. Results from an iterative closed-loop simulation campaign spanning multiple configurations demonstrate progressive convergence from initial instability to successful

simultaneous tracking of all three controlled variables across transient setpoint trajectories. The best configuration at $T_s = 0.01$ s achieves hydrogen blend ratio tracking within one percent steady-state error, generator load tracking within approximately half a megawatt, and turbine firing temperature offset of approximately six kelvins, with no fuel-control valve saturation during normal operation. A subsequent investigation re-identifies the discrete state-space model

at $T_s = 1.0$ s which is two orders of magnitude closer to the dominant plant time constants, and replaces the online Kalman covariance propagation with a fixed-gain observer derived from the offline discrete algebraic Riccati equation (DARE). Closed-loop simulation at the longer sample period confirms stable multi-variable control with physically meaningful prediction-matrix gains, while highlighting that model mismatch across a wide operating envelope remains the primary barrier to tracking accuracy at this time scale.

Quantum Algorithms for Patent Portfolio Analysis Using OpenModelica and IBM Qiskit

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Abstract

Near-term quantum computers are hybrid machines: classical computers, microprocessors, and control hardware drive a quantum subsystem that runs algorithms aimed at solving problems whose cost grows super-polynomially with input size. Prototype processors from research groups and commercial vendors are now accessible through software platforms such as IBM Qiskit, which offers both a circuit simulator (Qiskit Aer) and cloud access to physical quantum devices. Designing such classical-quantum hybrid systems can draw on the equation-based modeling and Hardware-in-the-Loop (HIL) techniques implemented in tools like OpenModelica. This paper integrates IBM Qiskit into the Modelica ecosystem through the OpenModelica external C interface and the Qiskit C API. We exercise the resulting solver on a 0/1 knapsack test case drawn from patent portfolio analysis as an example of asset management.

Keywords: Quantum computing, knapsack problem, patent portfolio

A Novel Modular Framework for Unified Data Center Cooling Virtual Testbed

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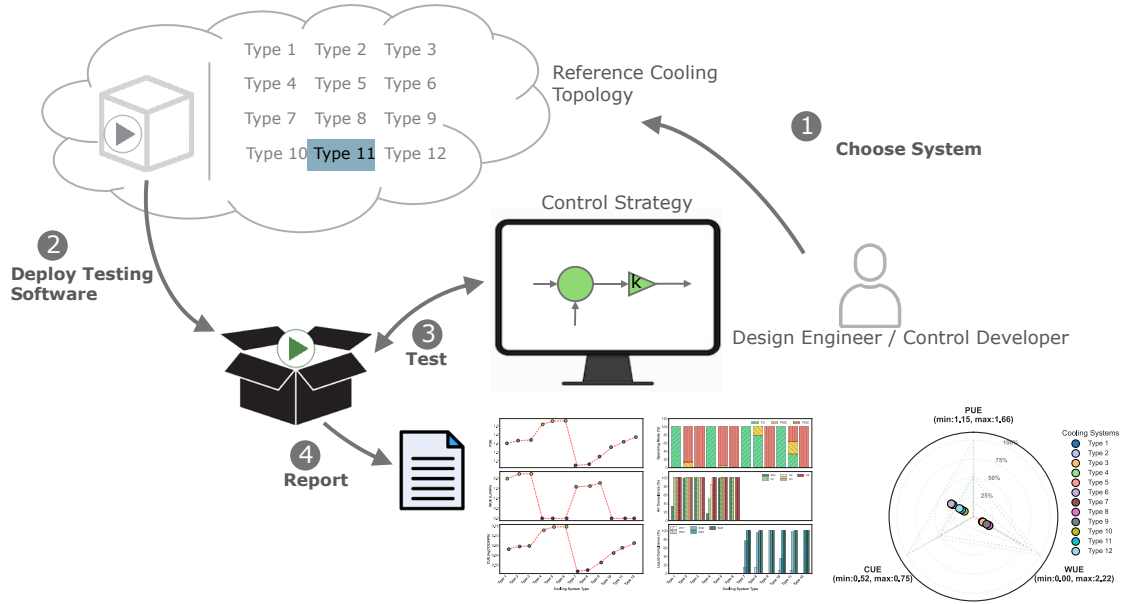


Figure 1. The concept (high level) (Ganesh et al. 2026).

Surging AI/HPC rack power densities (> 100 kW) render traditional, manual data center cooling workflows obsolete (Hoefler et al. 2022; Sankar et al. 2023). To address this, we present a Modelica-based virtual testbed and containerized Run-Time Environment (RTE) for rapid cooling co-design. Standardized control-bus interfaces enable modular integration of 12 cooling topologies from CRAHs to advanced liquid cooling (Heydari et al. 2022) with dynamic mode control. In this talk, attendees will be presented with a Climate Zone 1A case study showcasing automated control validation, thermal compliance, and multi-objective KPI tracking (PUE, WUE, CUE). The platform compresses design timelines from weeks to minutes; open challenges in hardware-in-the-loop (HIL) integration and hybrid optimization will also be discussed.

Keywords: Data Center Cooling, Virtual Test Bed, Modelica, Liquid Cooling, Air Cooling, Modular Framework

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Rapid De-risking and Sizing of GSHP Systems in Modelica

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Abstract

Continents, research and industry worlds collide in this research that informs the full design life cycle, from early stage to final design, by focusing on rapid de-risking, sizing, and value proposition creation related to ground source heat pump (GSHP) systems. Moreover, this collaboration showcases the value of pairing industry and research partners, united in a shared passion for open-source knowledge and best in class solutions.

This research explores the cumulative benefit of stacking system behaviors that are typically ignored in standard industry tools and quantifies the incremental value of each addition on GSHP systems. Examples include:

- 1) Borefield model that includes short-term thermal dynamics (impacts peak load response)
- 2) Accurate, full envelope performance of heat pump (impacts energy use and long-term thermal dynamics)
- 3) Dynamic fluid property calculation within the ground heat exchanger (*new MBL feature developed in this research*, impacts heat transfer to ground)
- 4) Horizontal piping distribution system volume and coupling to the ground (provides dynamic thermal storage, impacts peak load response)
- 5) System hydraulics, pump sizing, control, and heat gain to fluid

An early prototype of the Modelica system model is presented below.

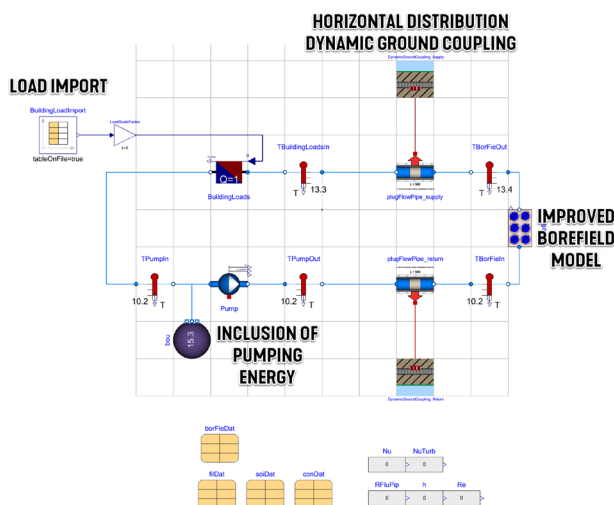


Figure 1 Modelica System Schematic of Base System (In-Progress)

The Modelica simulation results will be compared against data from recent projects designed utilizing typical industry tools across key performance metrics such as system temperature range and resultant ground heat exchanger size. Additional KPIs unique to the Modelica simulation will be included: system energy use, peak electrical power, hydraulic resistance and pump power.

A conceptual example set of results of ground heat exchanger (GHX) leaving temperature, a key metric in GHX sizing, is provided in the figure below. The baseline (Case 1) represents results from a traditional industry tool, while the other cases illustrate the possible incremental benefits of including the ignored behaviors via a Modelica-based simulation. Less extreme temperatures mean better heat pump performance and more abundant and affordable heat pump options.

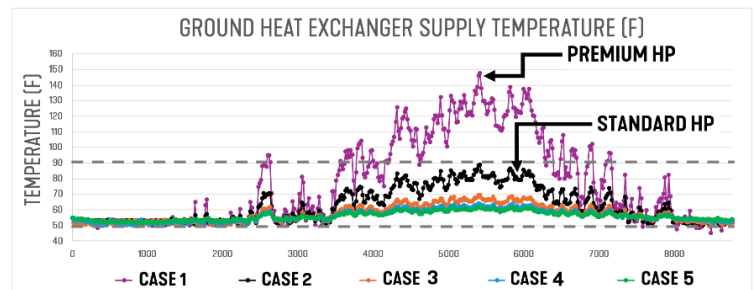


Figure 2 Conceptual results comparing the GHX Leaving Temperatures between a traditional sizing tool (Case 1) and a more accurate Modelica-based simulation (Cases 2-5).

Additionally, key benefits of Modelica modeling over traditional industry tools will be highlighted, notable advantages include:

- 1) Infrastructure de-risking benefits through transient response exploration, especially at small time steps under peak load conditions
- 2) Customizable open-source code
- 3) Expanded, multidomain system boundary
- 4) Inclusion of typically neglected system behavior
- 5) Ease of integration with AI models (for code debugging and refactoring)

Keywords: Thermal infrastructure de-risking and sizing, Ground source heat pump systems, Research and industry collaboration, Buildings Library in Modelon Impact

De-risking Energy System Integration through Platform-Based Design and Modelica

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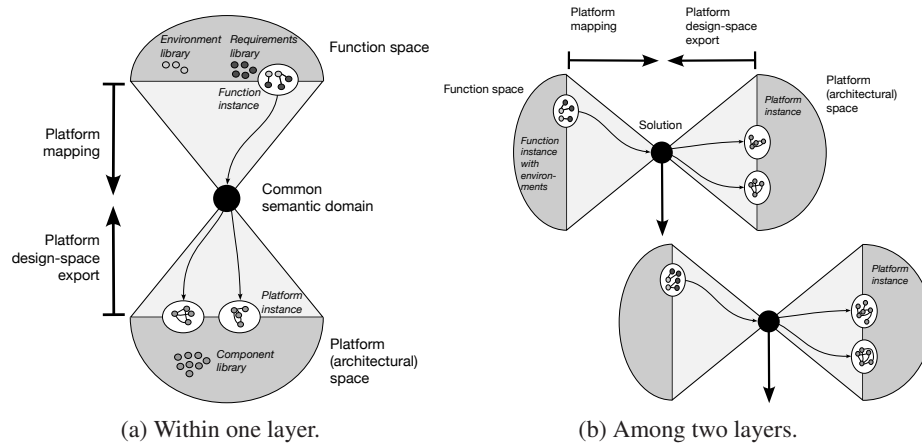


Figure 1. Platform-Based Design process within a platform layer, mapping functional requirements with architectural components to form a solution (a), and across two platform layers (b).

There is a significant market to retrofit heating and cooling systems for communities of buildings. For example, assuming a 25-year retrofit cycle, the US Department of Defense (DoD) would need to retrofit roughly two bases per month. To scale such retrofit projects, and to conduct them reliably and cost-effectively in a way that leads to high resilience and high energy supply security requires a shift from today's siloed waterfall practices, which delay integration checks until construction and turn system integration into a "plug and pray" endeavor, toward integrated planning and virtual testing.

Platform-Based Design (PBD), transferred from Electronic Design Automation, clearly separates functionality from implementation, and structures the design into discrete abstraction layers (Fig. 1). This enables holistic decision-making and de-risking of system integration, and allows for a high degree of reuse of proven components and subsystems, which is essential for scaling up retrofit projects.

We applied PBD to a 17-building retrofit at DoD's Joint Base Andrews (Wetter et al., 2026). MILP identified a geothermal Thermal Energy Network with photovoltaics and batteries as Pareto-optimal. For de-risking this system, we used the Modelica Buildings Library to build a fully coupled system of the geothermal borefield, the energy system and the controls.

De-risking of system integration with Modelica had a decisive impact. The MILP optimization was based on an assumed borefield heat transfer rate of 44 W/m; coupled simulation revealed 10 W/m for this installation, exposing an undersized central plant heat pump. Three rounds of control-logic and hydronic refinement followed, surfacing issues invisible in schematic reviews. Catching these pre-construction prevented change orders worth several million dollars, and sensitivity analyses identified first-cost savings of the same order, each offsetting the entire PBD cost. The finalized design reduces imported energy by 74% and peak demand by 45% vs. a gas-heat/air-source-chiller baseline.

Keywords: Platform-Based Design, Modelica Buildings Library, district energy, de-risking

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A Workflow for Enabling Modelica Building and District Models for DRTS-Ready FMUs

Nicholas Long Brian Ball Andy Olson Nischal Guruwacharya
Julian Osorio Rebecca Barney Whitney Trainor-Guitton

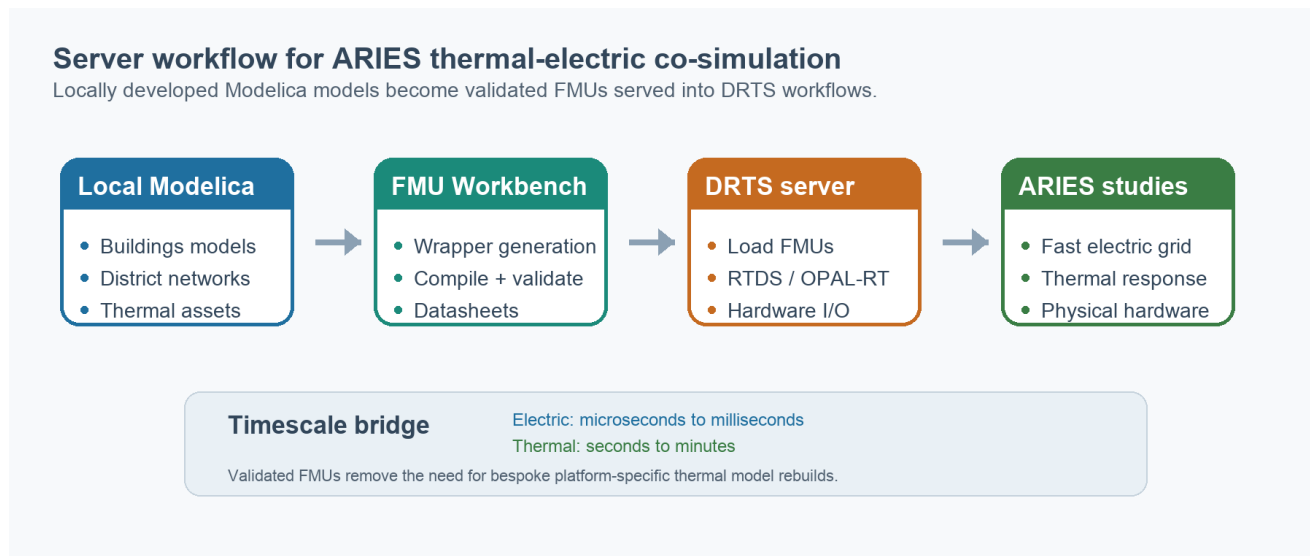


Figure 1. FMU workbench-to-DRTS server workflow for ARIES thermal-electric co-simulation.

The Advanced Research on Integrated Energy Systems (ARIES) platform enables studies that combine building and district thermal models, electrical systems, and physical hardware. These domains operate on different time scales: distribution-grid dynamics and power hardware-in-the-loop (HIL) studies may require microsecond-to-millisecond time steps, while heat pumps, hydronic loops, storage, ground coupling, district piping, and building loads respond over seconds to minutes. This presentation introduces a server workflow that converts locally developed Modelica thermal models into Digital Real-Time Simulation (DRTS)-ready Functional Mock-up Units (FMUs), supporting studies of thermal flexibility, data-center heat reuse, and district impacts on electric-grid stress.

The open-source FMU Compilation Workbench provides the translation layer shown in Figure 1. Acausal Modelica fluid connectors are converted into deterministic scalar input and output interfaces suitable for real-time platforms. For a selected component or topology, the workbench generates wrappers that bind media packages and expose signals such as mass-flow rate, inlet and outlet temperatures, heat flow, power, and coefficient of performance. It compiles the wrapped model with OpenModelica or Dymola as an FMI 2.0 Co-Simulation FMU, validates the result, and records deployment guidance. The component catalog includes heat pumps, boilers, coils, tanks, pumps, headers, district pipes, and buried pipe pairs.

FMU delivery is treated as a qualification process. Automated checks assess FMI structure, executable simulation, finite outputs, and physics consistency. The pipeline characterizes time-step sensitivity using empirical sweeps, step-response fitting, pseudo-random binary sequence bandwidth estimates, and multi-FMU coupling sweeps. These tests identify suitable communication steps, buffering requirements, and fragile coupling patterns. A DRTS server then loads validated FMUs and exchanges thermal signals with electrical models in Real Time Digital Simulator (RTDS) or OPAL-RT environments, preserving deterministic interaction between fast electrical dynamics and slower thermal responses without requiring bespoke thermal-model reconstruction in each platform.

The presentation will demonstrate the workflow and FMU usability for HIL applications, including initial results with buried pipe and heat-pump FMUs running alongside electrical systems on RTDS infrastructure. The approach combines interface design, server-side FMU management, and topology-level stability characterization so researchers can evaluate feeder voltage, power, building-load diversity, district temperatures, thermal storage, and hardware response together. It enables slow thermal flexibility to be studied in conjunction with fast electric-grid behavior and physical hardware.

Keywords: *Modelica, FMI, co-simulation, digital real-time simulation, hardware-in-the-loop, district energy, ARIES*

A Free-Open-Source Modelica Simulation Package for Air Conditioning and Refrigeration Cycle (SPARC)

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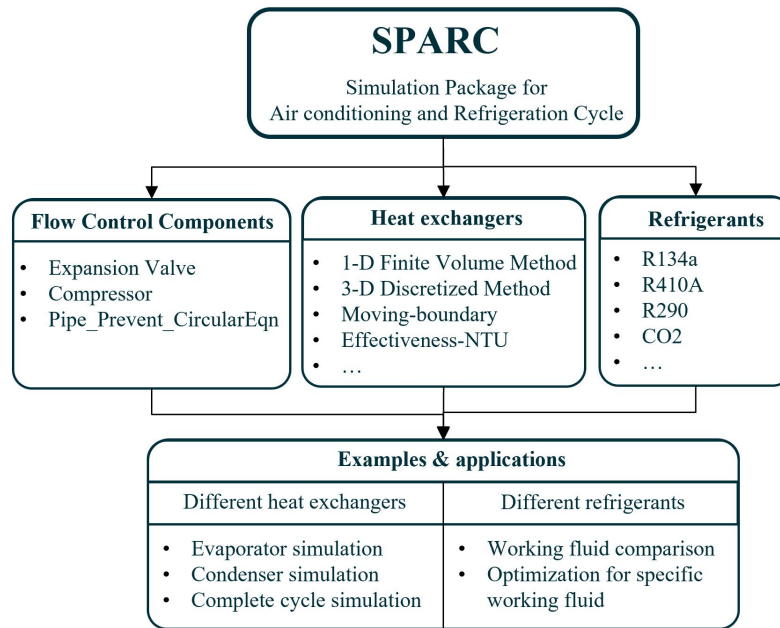


Figure 1. Overview of the SPARC package structure and its major component models.

Dynamic simulation of vapor-compression systems is increasingly important for system design, control, optimization, and fault detection. However, freely available and extensible tools for detailed refrigerant-cycle simulation remain limited. This presentation introduces the *Simulation Package for Air Conditioning and Refrigeration Cycle* (SPARC), a free and open-source Modelica framework for dynamic simulation of air-conditioning, refrigeration, and heat-pump systems, complementing existing Modelica-based HVAC modeling efforts (Li et al. 2014). SPARC is compatible with the Modelica Standard Library and the Buildings library (Wetter et al. 2014), facilitating integration with existing Modelica components and system models.

SPARC provides modular models for compressors, expansion valves, refrigerant properties, and heat exchangers using finite-volume, three-dimensional discretized, moving-boundary, and effectiveness-NTU formulations. The framework supports closed-cycle simulations and refrigerant-charge calculation. SPARC models can also be exported as Functional Mock-up Units (FMUs) for FMI-based co-simulation, performance analysis, and control optimization.

The finite-volume implementation is evaluated against experimental data from an R410A air-to-air heat pump, including 130 steady-state heating and cooling cases and a transient compressor-speed-change experiment. The simulations capture the major steady-state and transient behavior, with mean absolute percentage errors below 10% for most evaluated quantities. The presentation will introduce the SPARC architecture and modeling workflow, demonstrate system-level applications, summarize experimental validation, and discuss current limitations and ongoing development.

Keywords: *Modelica, open source, vapor compression cycle, heat pump, refrigeration, FMI*

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Real-Code Models for Electromagnetic Transient Simulations: Interoperability with Modelica

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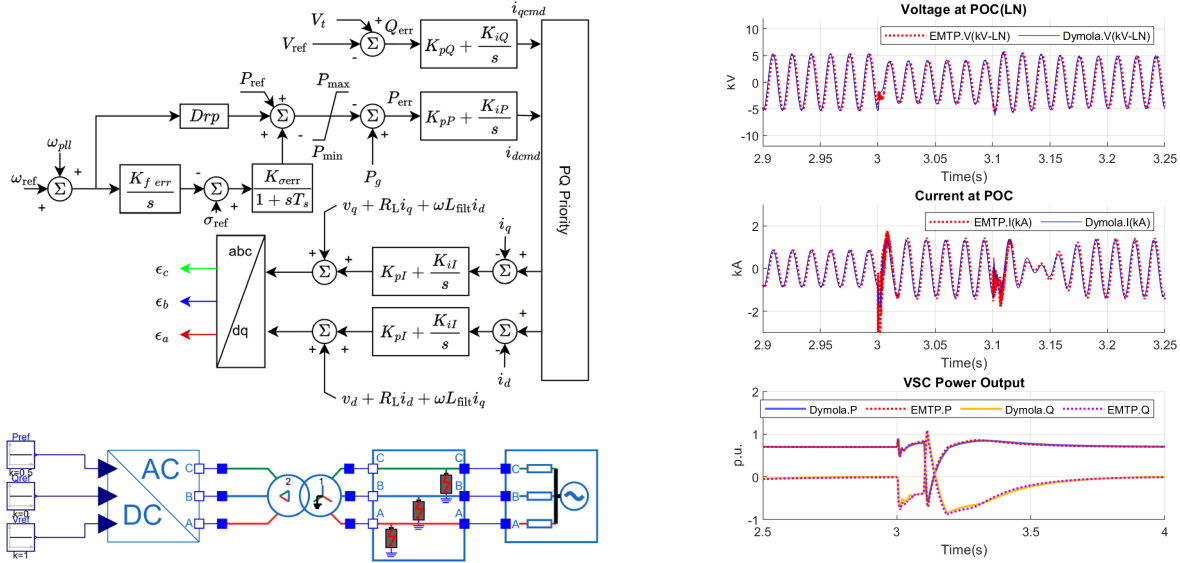


Figure 1. GFL controller and EMT network (left); Modelica and EMTP[®] fault responses (right) (Chang and Vanfretti 2026b).

Vendor-supplied power system controller models are increasingly distributed as compiled shared libraries (DLLs). IEEE/CIGRE Technical Brochure 958 defines a common C-language DLL interface for embedding these real-code models in electromagnetic transient (EMT) tools (CIGRE JWG B4.82 / IEEE 2025), but their use remains concentrated in proprietary simulators. This presentation demonstrates how Modelica’s `ExternalObject` construct bridges industry-standard controller DLLs and the Modelica ecosystem, retaining validated controller code within broader modeling and simulation workflows.

The interface uses a constructor to initialize a DLL instance and a destructor to release it, linked through external C functions. Physical parameters are exposed through the constructor and remain tunable in Modelica. Earlier work demonstrated this approach for a positive-sequence power system model with a static excitation controller (Chang and Vanfretti 2024). The EMT case introduces a synchronization challenge: the controller firmware executes at a fixed 50 μ s time step, while the surrounding Modelica model uses a variable-step continuous-time solver. A C wrapper samples the solver state and advances the firmware in fixed ticks at each communication point.

A grid-following voltage-source converter controller DLL is integrated into a Modelica EMT network using Dymola. Figure 1 shows the controller, network, and three-phase fault response, with close agreement between the Modelica-integrated DLL and an EMTP[®] reference (Chang and Vanfretti 2026b). Remaining differences are attributed to the surrounding Modelica EMT component models, which require further validation. The presentation discusses implementation, synchronization, and validation, including future use of clocked partitions and synchronous language features. Complementary applications include FMI model exchange for commercial simulator deployment and eFMI for traceable embedded-code generation and controller hardware-in-the-loop testing.

Keywords: *Modelica, ExternalObject, DLL, real-code model, power systems, interoperability, EMT*

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Optimization of a Green Hydrogen Production Case Study Utilizing a Sigmoid Function

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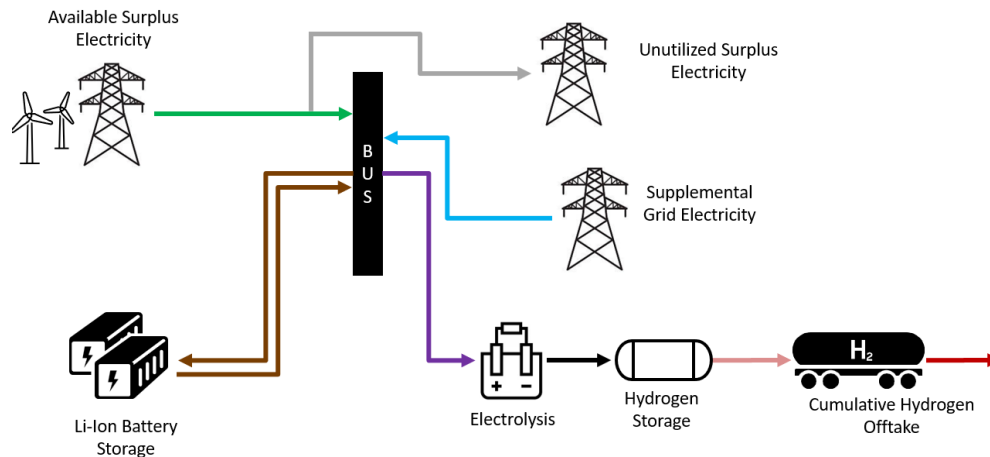


Figure 1. Model Configuration Schematic.

Hydrogen holds significant potential to aid in economy-wide decarbonization. However, regional factors such as access to renewable power generation, hydrogen offtake demand, and most importantly, the cost of electricity and the cost of hydrogen, can introduce many uncertainties into the feasibility of deploying electrolysis hydrogen production plants. In recent years, EPRI's Low-Carbon Resources Initiative (LCRI) has developed a Hydrogen Electrolysis Technoeconomic Analysis Tool that provides an interface to calculate high-level overviews of hydrogen cost based on user-input parameters [1]. This tool along with the National Laboratory of the Rockies (NLR)'s System Advisor Model (SAM) were utilized within previous EPRI-published research focused on several solar + electrolysis and wind + electrolysis case studies to evaluate levelized cost of hydrogen (LCOH) for various climate regions [2][3].

Recent EPRI research has focused on utilizing Modelica-based tools such as Modelon Impact to develop hybrid energy system configurations for dynamic simulations and optimization. This allows each case study and/or sensitivity to be optimized based on several constraints as well as production and economic targets. Typical EPRI-led green hydrogen optimization studies have focused on maximizing renewables and battery electricity to optimize the capacity factor of the electrolyzer in an effort to reduce LCOH and total cost of ownership (TCO). However, this work highlights the development of an optimized case study simulation focused on delivering green hydrogen only during periods when surplus renewable electricity is unavailable. Contrastingly, this model's controls also dictate that hydrogen production should only be produced during periods when renewables and/or grid electricity are economically advantageous, with a prioritization towards green hydrogen. As a result, a customized Sigmoid function was required to access the surplus renewables electricity profile and adequately ramp hydrogen off-take, while also maintaining optimization functionalities to determine optimal capacities and operational profiles of the electrolyzer, battery storage, compressor, and hydrogen storage components. This customization was incorporated into Modelon Impact via customized Modelica language. Case study parameters and initial results of this optimized case study will be presented, along with a summarization of the Sigmoid function setup within Modelon Impact.

Keywords: User Presentation, Optimization, Modelica, Sigmoid Function, Green Hydrogen, Energy Systems

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Simulating Renewable Load-Following Hydrogen Electrolysis Plant Controls

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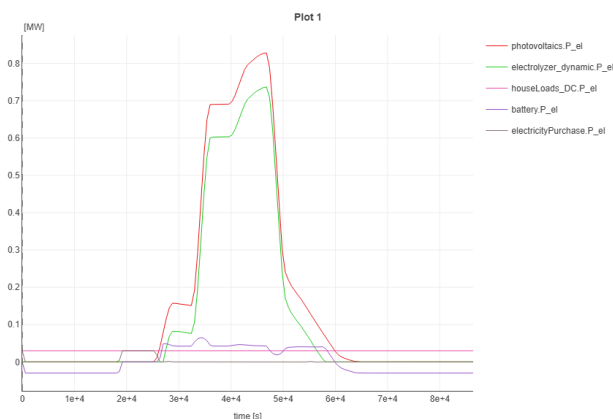


Figure 1. Power profiles from simulated hydrogen microgrid.

This study evaluates control strategies for a green hydrogen facility under different operation scenarios. Both islanded and grid-connected scenarios were run, evaluating the ability of renewable power plus battery storage to regularly run the electrolyzer and handle other hydrogen balance of plant loads for extended periods of time. All controls were developed with Modelica library components, and the model was developed using Modelon Impact.

EPRI is working with various research groups, utilizing real and simulated solar and wind data from potential electrolysis sites, and has developed models to evaluate initial performance estimates of hydrogen systems at these sites. This work is a continuation of the study EPRI presented at the 16th International Modelica & FMI Conference conference comparing PEM and alkaline electrolyzer technology performance metrics and operational constraints. Utilizing renewable energy profiles and customized electrolyzer components, control strategies were developed to maximize electrolyzer operation and minimize reliance on supplemental power from the grid. These constraints can be enforced or relaxed depending on project parameters and highlight potential advantages and disadvantages of having additional equipment, such as batteries, on site for these projects.

Multiple plant control approaches are under development that allow EPRI to evaluate varying objectives and constraints across projects. Some controls ensure that the renewable profiles provided are sufficient to handle all loads at the site. Others allow some grid electricity to be purchased as needed. Control strategies implemented are primarily self-regulating process, integrating process, and feedforward control approaches [1].

Through the controls logic implemented in this model, a 1-MW electrolyzer was able to operate in accordance with the available solar and supplemental battery power with minimal electricity purchase from the grid. Though there is limited publicly available data today on electrolyzer degradation, and degradation mechanisms will vary by electrolyzer technology type, compositions of membranes, electrolytes, and other factors pertaining to the electrochemical reaction, this model could be useful for evaluating how electrolyzer operation may impact degradation in future systems. By characterizing and quantifying ramp events, or times when the electrolyzer system is asked to ramp up or down faster than certain critical components within it are truly capable of, assumptions could be made on how those events may impact degradation of the system. Future work will be focused on setting some of those assumptions and creating degradation models to test in this simulation.

Keywords: *Electrolyzer, Green hydrogen, Battery storage, Plant controls, Load-following*

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VENDOR ABSTRACTS





JuliaHub: Agentic Modeling and Simulation with Dyad

JuliaHub is defining agentic engineering with Dyad, an AI-native platform for modeling, simulation, and control-system development. Dyad combines AI agents with a physics-based, equation-oriented modeling language and the high-performance Julia and SciML ecosystem. Engineers can describe a system in natural language, import existing requirements, and use the Dyad Agent to build, simulate, troubleshoot, and refine executable models. A compiler checks the agent's work for unit consistency, connection integrity, and physical validity, helping teams move faster without sacrificing engineering rigor.

Dyad unifies physical models, control logic, data-driven methods, and production code in a single, traceable workflow. It supports applications across aerospace, automotive, HVAC, energy, utilities, and other complex software-defined systems. By integrating with established engineering workflows and supporting model exchange through standards such as FMI, Dyad helps organizations explore more designs, identify problems earlier, and turn engineering knowledge into reusable, validated models.

JuliaHub will present the Agentic Modeling and Simulation with Julia/Dyad workshop at the American Modelica & FMI Conference, organized by NAMUG in cooperation with the Modelica Association.



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You will discover the latest news and perspectives regarding Dymola and underlying technologies, as well as the portfolio of libraries. Also included: latest standards support and examples of workflows involving native and web clients of the Dassault Systèmes offer. Highlights of the vendor session include:

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- Improvements for co-simulation of FMUs.
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Built by Modelon, a long-time leader in Modelica-based engineering software, Modelon Impact is founded on open standards including Modelica and FMI. Organizations can protect existing investments, integrate with broader engineering ecosystems, and increase the value of simulation across the product development process.

Learn more at our presentations and workshop, “Physics + AI in Practice: Accelerating Modelica Workflows with Modelon Impact.” Discover how agentic AI can reduce workflow friction, accelerate troubleshooting, and make Modelica-based engineering more productive without sacrificing engineering rigor.

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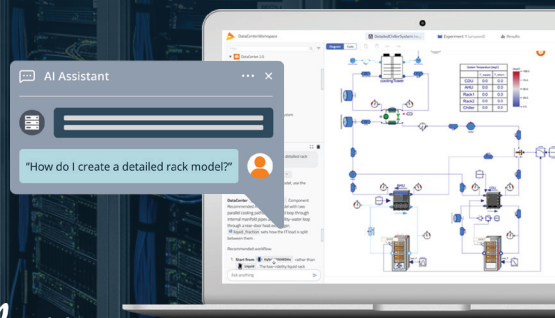
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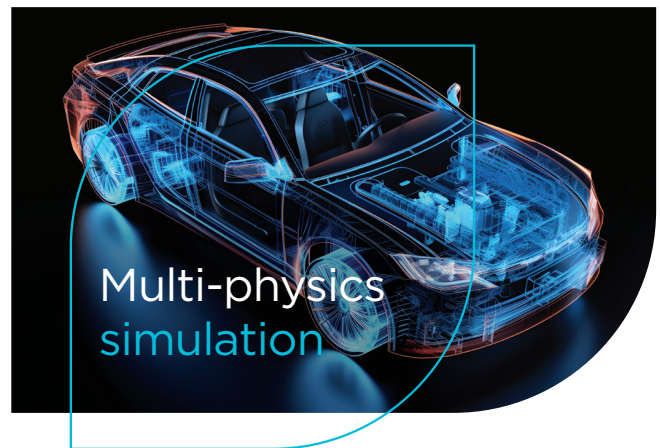

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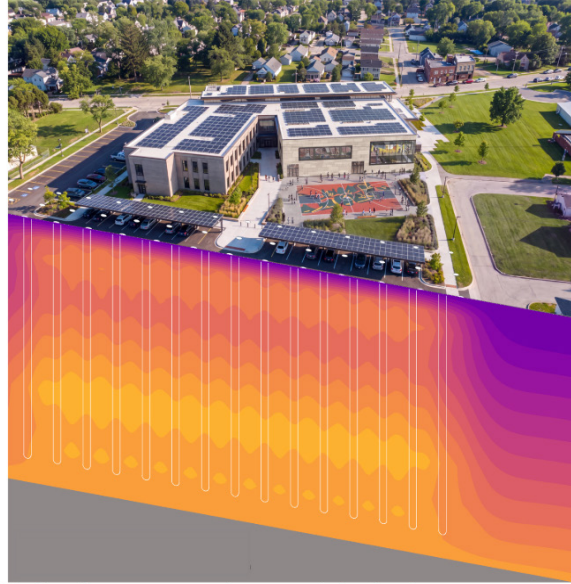
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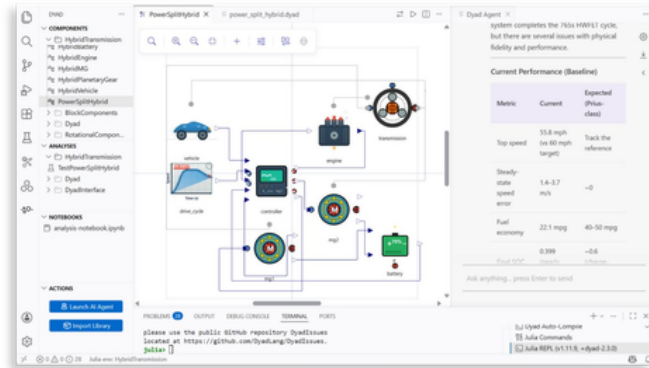
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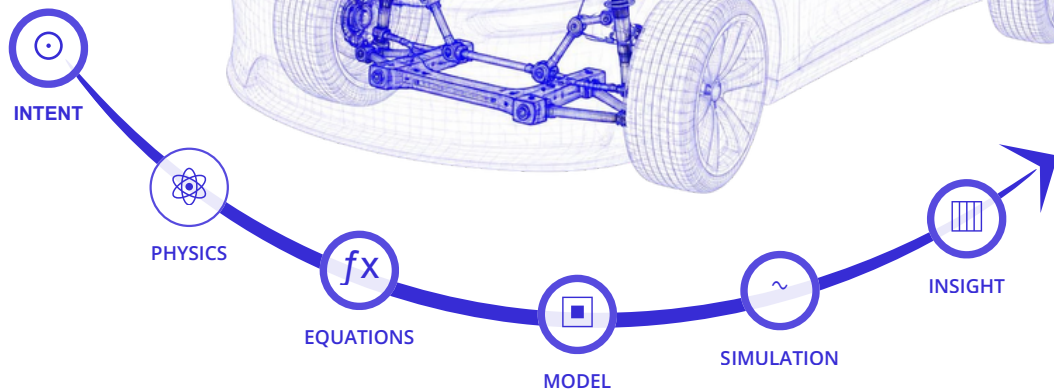


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